NAMA PEMILIK PROYEK : **PT PERTAMINA EP**

NAMA PROYEK : **JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR**

LOKASI : **SP TANJUNG MIRING BARAT & KARANG TIMUR - FIELD LIMAU**

NO. KONTRAK : **SA02035383B**

REV.	TGL.	DESKRIPSI	DISIAPKAN	DIPERIKSA	DISETUJUI	DIPERIKSA	DISETUJUI
A	07/01/2026	Issued For Review	BS	AS	RTN		
			PT. PROSYMPAC OIL & GAS			PT. PERTAMINA EP	

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Rev.	Tanggal	Deskripsi

COMMENT RESPONSE SHEET

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1. *GENERAL*

1.1 Latar Belakang

Struktur Tanjung Miring Barat (TMB) terletak di Kabupaten Muara Enim dan Kota Prabumulih, sekitar 20 kilometer di sebelah tenggara Kota Prabumulih, dengan luas daerah sekitar 8 km². Struktur Karangan (KRG) terletak di Kabupaten Muara Enim dan Kota Prabumulih, sekitar 15 kilometer di sebelah selatan Kota Prabumulih, dengan luas daerah sekitar 45 km².

ID	Easting (Meter)	Northing (Meter)	Elevation (Meter)	Feature Code
PN-1531	424962,745	967376,210	10,414	
PM-8593	425000,093	907528,777	19,046	

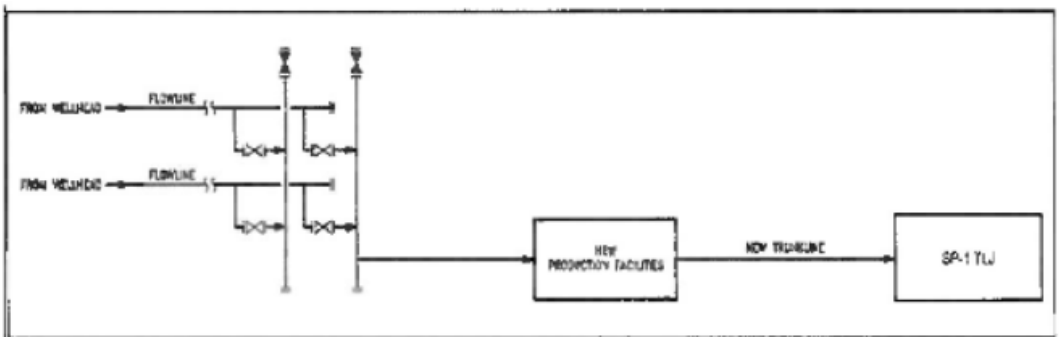
Gambar 1. Lokasi Rencana SP Tanjung Miring Barat - Karangan Timur

Lapangan Tanjung Miring Barat telah diproduksi sejak tahun 1938 dan terus beroperasi hingga tahun 2003 dengan produksi terakhir dari sumur TMB-11. Sejak dinonaktifkannya lapangan ini, fasilitas produksi yang terdiri dari *flowline*, stasiun pengumpul, serta jalur pengiriman pipa minyak ke *custody transfer* (PPP Prabumulih) sudah tidak layak operasi. Apabila akan dilakukan pengembangan lapangan, diperlukan pembangunan fasilitas produksi baru yang dapat menampung produksi hidrokarbon dari lapangan Tanjung Miring Baral serta memprosesnya agar sesuai dengan spesifikasi yang dipersyaratkan di titik serah. Saat ini status SP TMB adalah *idle* atau tidak beroperasi, sehingga produksi sumuran menggunakan *Tank On-Site* (TOS) dan dikirim secara *gross* ke SP-II TMT menggunakan *trucking*. Dikarenakan tidak adanya *line gas* ke sumur-sumur struktur TMB, *supply* listrik untuk *pumping* unit menggunakan solar *(diesel engine)*. Lapangan Karangan Timur hanya diproduksikan dari 2 sumur yaitu KRG-9 dan KRG-10 beroperasi mulai tahun 1984, dimana kondisi eksisting jalur pengiriman minyak dari sumur menggunakan *trucking*. Apabila akan dilakukan pengembangan lapangan, diperlukan pembangunan fasilitas produksi baru yang dapat menampung produksi

hidrokarbon dari lapangan Tanjung Miring Barat serta memprosesnya agar sesuai dengan spesifikasi yang dipersyaratkan di titik serah.

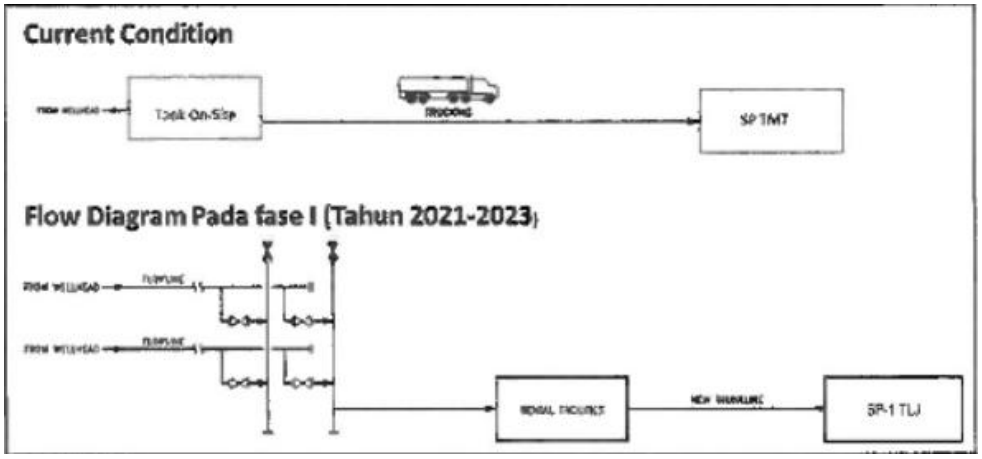
Dalam rencana jangka panjang sebagai upaya peningkatan produksi, PT Pertamina EP berencana akan mengembangkan Lapangan Tanjung Miring Barat dan Karangan Timur melalui pengembangan fasilitas produksi dimana rencana kerja pengembangan fasilitas akan dibagi ke dalam dua fase.

Pada fase pertama, produksi dari pengembangan awal TMB -KRG Timur akan dikirimkan menuju fasilitas sewa *Early Production Facility* (EPF) untuk selanjutnya dikirimkan menuju SP-1 TLJ melalui trunkline baru. Pada fase ini lingkup fasilitas produksi yang akan dibangun adalah *flowline* dari sumuran, sewa EPF dan *trunkline* dari EPF sampai dengan SP-1 TLJ.



Gambar 2. Rencana Pola Operasi Fase

Pada fase kedua, akan dilakukan pengembangan tambahan dari sumur-sumur TMB-KRG Timur. Pada fase ini, lingkup fasilitas produksi yang akan dibangun meliputi pembangunan *flowline* dari sumuran pengembangan lanjutan serta peningkatan fasilitas SP TMB untuk menggantikan sewa EPF.



Gambar 3. Rencana Pola Operasi Fase

1.2 Tujuan

Tujuan dari dokumen ini adalah memberikan informasi untuk *Bill of Material* yang digunakan berdasarkan *design* pada proyek Jasa Pembangunan Fasilitas Produksi Stasiun Pengumpul (SP) Tanjung Miring Barat & Karangan Timur, Kabupaten Muara Enim dan Kota Prabumulih.

1.3 Ruang Lingkup

Ruang lingkup dalam dokumen ini meliputi *Bill of Material* untuk item *Pipe, Flange & Fitting*.

1.4 Istilah

Dokumen ini berisi istilah-istilah yang memiliki definisi tercantum sebagai berikut:

Perusahaan	Perusahaan yang dimaksud adalah PT PERTAMINA HULU ROKAN selaku Pemilik dan Pengguna Fasilitas akhir.
Kontraktor	Pihak yang dipilih oleh PERUSAHAAN sebagai pelaksana pekerjaan pengadaan dan konstruksi yang akan bertanggung jawab kepada PERUSAHAAN.
Vendor	Pihak lain yang akan menyediakan sesuatu untuk kebutuhan proyek, selain yang disediakan oleh PERUSAHAAN dan KONTRAKTOR, disebut VENDOR. Tanggung jawabnya langsung kepada KONTRAKTOR baik terhadap PO maupun bentuk perjanjian lain antara VENDOR dan KONTRAKTOR.
SKK MIGAS	Suatu badan dari Pemerintah Indonesia yang bertanggung jawab terhadap proses persetujuan dan izin untuk semua fasilitas di lingkungan minyak dan gas.
Perencana	Pihak yang dipilih oleh PERUSAHAAN sebagai pelaksana pekerjaan Jasa FEED Engineering Services. Dalam hal ini disebut PERENCANA, yang akan bertanggung jawab kepada PERUSAHAAN sebagaimana dirinci dalam kontrak, scope of work, atau work order.
Inspektor	Perwakilan dari PERUSAHAAN dan/atau seseorang yang ditunjuk oleh PERUSAHAAN, yang bertanggung jawab sebagai INSPEKTOR atas pekerjaan yang dilakukan oleh KONTRAKTOR.SINGKATAN

ATTACHMENT-1 DETAIL MTO PIPES, FLANGES AND FITTINGS PER LINE NUMBER

DETAIL MTO PIPES, FLANGES AND FITTINGS (AS PER LINE NUMBER)

JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR

NO.	AREA	LINE NUMBER	PID REFERENCE	PART NAME	MATERIAL CLASS	SCHEDULE	RATING	CONNECTION END	DESCRIPTION	SIZE (Inch)	QTY	UNIT	Remarks
1	AREA PRODUKSI	DC-001-AA-4"	TMBP-01-PR-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	93,042	M	
2	AREA PRODUKSI	DC-001-AA-4"	TMBP-01-PR-PID-101	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
3	AREA PRODUKSI	DC-001-AA-4"	TMBP-01-PR-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
4	AREA PRODUKSI	DC-001-AA-4"	TMBP-01-PR-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
5	AREA PRODUKSI	DC-001-AA-4"	TMBP-01-PR-PID-101	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
6	AREA PRODUKSI	DC-001-AA-4"	TMBP-01-PR-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
7	AREA PRODUKSI	DC-001-AA-4"	TMBP-01-PR-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
8	AREA PRODUKSI	DC-001-AA-4"	TMBP-01-PR-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
9	AREA PRODUKSI	DC-003-AA-4"	TMBP-01-PR-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	51,284	M	
10	AREA PRODUKSI	DC-003-AA-4"	TMBP-01-PR-PID-101	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
11	AREA PRODUKSI	DC-003-AA-4"	TMBP-01-PR-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
12	AREA PRODUKSI	DC-003-AA-4"	TMBP-01-PR-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
13	AREA PRODUKSI	DC-004-AA-2"	TMBP-01-PR-PID-102	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
14	AREA PRODUKSI	DC-004-AA-2"	TMBP-01-PR-PID-102	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
15	AREA PRODUKSI	DC-004-AA-2"	TMBP-01-PR-PID-102	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
16	AREA PRODUKSI	DC-004-AA-2"	TMBP-01-PR-PID-102	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	3	EA	
17	AREA PRODUKSI	DC-004-AA-2"	TMBP-01-PR-PID-102	SPECTACLE BLIND	AA		150	RF	SPECTACLE BLIND, A515-70, CL.150, RF, ASME B16.48	4"	1	EA	
18	AREA UTILITY	DC-009-AA-2"	TMBP-01-PR-PID-118	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	7,532	M	
19	AREA UTILITY	DC-009-AA-2"	TMBP-01-PR-PID-118	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
20	AREA UTILITY	DC-009-AA-2"	TMBP-01-PR-PID-118	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
21	AREA UTILITY	DC-009-AA-2"	TMBP-01-PR-PID-118	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
22	AREA UTILITY	DC-009-AA-2"	TMBP-01-PR-PID-118	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
23	AREA UTILITY	DC-029-AA-4"	TMBP-01-PR-PID-116	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	150,368	M	
24	AREA UTILITY	DC-029-AA-4"	TMBP-01-PR-PID-116	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
25	AREA UTILITY	DC-029-AA-4"	TMBP-01-PR-PID-116	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
26	AREA UTILITY	DC-029-AA-4"	TMBP-01-PR-PID-116	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	3	EA	
27	AREA UTILITY	DC-029-AA-4"	TMBP-01-PR-PID-116	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
28	AREA UTILITY	DC-029-AA-4"	TMBP-01-PR-PID-116	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	5	EA	
29	AREA UTILITY	DC-029-AA-4"	TMBP-01-PR-PID-116	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
30	AREA UTILITY	DC-029-AA-4"	TMBP-01-PR-PID-116	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	3	EA	
31	AREA UTILITY	DC-029-AA-4"	TMBP-01-PR-PID-116	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
32	AREA UTILITY	DC-029-AA-4"	TMBP-01-PR-PID-116	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
33	AREA UTILITY	DC-029-AA-4"	TMBP-01-PR-PID-116	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
34	AREA UTILITY	DC-029-AA-4"	TMBP-01-PR-PID-116	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	32	EA	
35	AREA UTILITY	DC-029-AA-4"	TMBP-01-PR-PID-116	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	4	EA	
36	AREA PRODUKSI	DC-065-AA-2"	TMBP-01-PR-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
37	AREA PRODUKSI	DO-001-AA-4"	TMBP-01-PR-PID-103	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	15,746	M	
38	AREA PRODUKSI	DO-001-AA-4"	TMBP-01-PR-PID-103	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
39	AREA PRODUKSI	DO-001-AA-4"	TMBP-01-PR-PID-103	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
40	AREA PRODUKSI	DO-001-AA-4"	TMBP-01-PR-PID-103	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
41	AREA PRODUKSI	DO-001-AA-4"	TMBP-01-PR-PID-103	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
42	AREA PRODUKSI	DO-001-AA-4"	TMBP-01-PR-PID-103	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
43	AREA PRODUKSI	DO-001-AA-4"	TMBP-01-PR-PID-103	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
44	AREA UTILITY	DO-008-AA-4"	TMBP-01-PR-PID-118	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	12,853	M	
45	AREA UTILITY	DO-008-AA-4"	TMBP-01-PR-PID-118	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
46	AREA UTILITY	DO-008-AA-4"	TMBP-01-PR-PID-118	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
47	AREA UTILITY	DO-008-AA-4"	TMBP-01-PR-PID-118	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
48	AREA UTILITY	DO-008-AA-4"	TMBP-01-PR-PID-118	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
49	AREA UTILITY	DO-008-AA-4"	TMBP-01-PR-PID-118	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
50	AREA UTILITY	DO-008-AA-4"	TMBP-01-PR-PID-118	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
51	AREA UTILITY	DO-008-AA-4"	TMBP-01-PR-PID-118	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
52	AREA UTILITY	DO-009-AA-4"	TMBP-01-PR-PID-118	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	7,572	M	
53	AREA UTILITY	DO-009-AA-4"	TMBP-01-PR-PID-118	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
54	AREA UTILITY	DO-009-AA-4"	TMBP-01-PR-PID-118	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
55	AREA UTILITY	DO-009-AA-4"	TMBP-01-PR-PID-118	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
56	AREA UTILITY	DO-009-AA-4"	TMBP-01-PR-PID-118	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
57	AREA UTILITY	DO-009-AA-4"	TMBP-01-PR-PID-118	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
58	AREA UTILITY	DO-009-AA-4"	TMBP-01-PR-PID-118	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
59	AREA PRODUKSI	DO-019-AA-4"	TMBP-01-PR-PID-106	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	24,1	M	
60	AREA PRODUKSI	DO-019-AA-4"	TMBP-01-PR-PID-106	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
61	AREA PRODUKSI	DO-019-AA-4"	TMBP-01-PR-PID-106	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
62	AREA PRODUKSI	DO-019-AA-4"	TMBP-01-PR-PID-106	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
63	AREA PRODUKSI	DO-019-AA-4"	TMBP-01-PR-PID-106	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
64	AREA PRODUKSI	DO-019-AA-4"	TMBP-01-PR-PID-106	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
65	AREA PRODUKSI	DO-019-AA-4"	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
66	AREA PRODUKSI	DO-019-AA-4"	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
67	AREA PRODUKSI	DO-020-AA-4"	TMBP-01-PR-PID-106	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	2,46	M	
68	AREA PRODUKSI	DO-020-AA-4"	TMBP-01-PR-PID-106	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
69	AREA PRODUKSI	DO-020-AA-4"	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
70	AREA PRODUKSI	DO-020-AA-4"	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
71	AREA PRODUKSI	DO-023-AA-6"	TMBP-01-PR-PID-106	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	46,413	M	
72	AREA PRODUKSI	DO-023-AA-6"	TMBP-01-PR-PID-106	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	0,273	M	
73	AREA PRODUKSI	DO-023-AA-6"	TMBP-01-PR-PID-106	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
74	AREA PRODUKSI	DO-023-AA-6"	TMBP-01-PR-PID-106	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	3	EA	
75	AREA PRODUKSI	DO-023-AA-6"	TMBP-01-PR-PID-106	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
76	AREA PRODUKSI	DO-023-AA-6"	TMBP-01-PR-PID-106	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	2	EA	
77	AREA PRODUKSI	DO-023-AA-6"	TMBP-01-PR-PID-106	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
78	AREA PRODUKSI	DO-023-AA-6"	TMBP-01-PR-PID-106	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
79	AREA PRODUKSI	DO-023-AA-6"	TMBP-01-PR-PID-106	REDUCING TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	2	EA	
80	AREA PRODUKSI	DO-023-AA-6"	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	16	EA	
81	AREA PRODUKSI	DO-023-AA-6"	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
82	AREA PRODUKSI	DO-023-AA-6"	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	2	EA	
83	AREA PRODUKSI	DO-023-AA-6"	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
84	AREA UTILITY	DO-041-AA-4"	TMBP-01-PR-PID-117	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	17,929	M	
85	AREA UTILITY	DO-041-AA-4"	TMBP-01-PR-PID-117	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
86	AREA UTILITY	DO-041-AA-4"	TMBP-01-PR-PID-117	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
87	AREA UTILITY	DO-041-AA-4"	TMBP-01-PR-PID-117	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
88	AREA UTILITY	DO-041-AA-4"	TMBP-01-PR-PID-117	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
89	AREA UTILITY	DO-041-AA-4"	TMBP-01-PR-PID-117	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
90	AREA UTILITY	DO-041-AA-4"	TMBP-01-PR-PID-117	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
91	AREA UTILITY	DO-041-AA-4"	TMBP-01-PR-PID-117	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	24	EA	
92	AREA UTILITY	DO-041-AA-4"	TMBP-01-PR-PID-117	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	3	EA	
93	AREA PRODUKSI	DO-042-AA-4"	TMBP-01-PR-PID-114	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	4,857	M	
94	AREA PRODUKSI	DO-042-AA-4"	TMBP-01-PR-PID-114	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
95	AREA PRODUKSI	DO-042-AA-4"	TMBP-01-PR-PID-114	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
96	AREA PRODUKSI	DO-042-AA-4"	TMBP-01-PR-PID-114	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
97	AREA PRODUKSI	DO-042-AA-4"	TMBP-01-PR-PID-114	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
98	AREA UTILITY	DO-044-AA-4"	TMBP-01-PR-PID-117	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	16,014	M	
99	AREA UTILITY	DO-044-AA-4"	TMBP-01-PR-PID-117	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
100	AREA UTILITY	DO-044-AA-4"	TMBP-01-PR-PID-117	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
101	AREA UTILITY	DO-044-AA-4"	TMBP-01-PR-PID-117	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
102	AREA UTILITY	DO-044-AA-4"	TMBP-01-PR-PID-117	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
103	AREA UTILITY	DO-044-AA-4"	TMBP-01-PR-PID-117	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
104	AREA UTILITY	DO-044-AA-4"	TMBP-01-PR-PID-117	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
105	AREA PRODUKSI	DO-045-AA-4"	TMBP-01-PR-PID-104	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	1,851	M	
106	AREA PRODUKSI	DO-045-AA-4"	TMBP-01-PR-PID-104	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
107	AREA PRODUKSI	DO-045-AA-4"	TMBP-01-PR-PID-104	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
108	AREA PRODUKSI	DO-045-AA-4"	TMBP-01-PR-PID-104	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
109	AREA PRODUKSI	DO-045-AA-4"	TMBP-01-PR-PID-104	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
110	AREA PRODUKSI	DO-047-AA-4"	TMBP-01-PR-PID-106	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	5,727	M	
111	AREA PRODUKSI	DO-047-AA-4"	TMBP-01-PR-PID-106	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	2,044	M	
112	AREA PRODUKSI	DO-047-AA-4"	TMBP-01-PR-PID-106	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
113	AREA PRODUKSI	DO-047-AA-4"	TMBP-01-PR-PID-106	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
114	AREA PRODUKSI	DO-047-AA-4"	TMBP-01-PR-PID-106	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	4	EA	
115	AREA PRODUKSI	DO-047-AA-4"	TMBP-01-PR-PID-106	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
116	AREA PRODUKSI	DO-047-AA-4"	TMBP-01-PR-PID-106	REDUCING TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	1	EA	
117	AREA PRODUKSI	DO-047-AA-4"	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	32	EA	
118	AREA PRODUKSI	DO-047-AA-4"	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	4	EA	
119	AREA PRODUKSI	DO-048-AA-4"	TMBP-01-PR-PID-106	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	5,646	M	
120	AREA PRODUKSI	DO-048-AA-4"	TMBP-01-PR-PID-106	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
121	AREA PRODUKSI	DO-048-AA-4"	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	

DETAIL MTO PIPES, FLANGES AND FITTINGS (AS PER LINE NUMBER)

JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR

NO.	AREA	LINE NUMBER	PID REFERENCE	PART NAME	MATERIAL CLASS	SCHEDULE	RATING	CONNECTION END	DESCRIPTION	SIZE (Inch)	QTY	UNIT	Remarks
122	AREA PRODUKSI	DO-048-AA-4"	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
123	AREA PRODUKSI	DO-050-AA-4"	TMBP-01-PR-PID-105	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	55,608	M	
124	AREA PRODUKSI	DO-050-AA-4"	TMBP-01-PR-PID-105	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	7	EA	
125	AREA PRODUKSI	DO-050-AA-4"	TMBP-01-PR-PID-105	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	9	EA	
126	AREA PRODUKSI	DO-050-AA-4"	TMBP-01-PR-PID-105	CONCENTRIC REDUCER	AA	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	5	EA	
127	AREA PRODUKSI	DO-050-AA-4"	TMBP-01-PR-PID-105	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL 150, RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
128	AREA PRODUKSI	DO-050-AA-4"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
129	AREA PRODUKSI	DO-050-AA-4"	TMBP-01-PR-PID-105	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	5	EA	
130	AREA PRODUKSI	DO-050-AA-4"	TMBP-01-PR-PID-105	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
131	AREA PRODUKSI	DO-050-AA-4"	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
132	AREA UTILITY	DO-051-AA-4"	TMBP-01-PR-PID-119	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	2,526	M	
133	AREA UTILITY	DO-051-AA-4"	TMBP-01-PR-PID-119	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
134	AREA UTILITY	DO-051-AA-4"	TMBP-01-PR-PID-119	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
135	AREA UTILITY	DO-051-AA-4"	TMBP-01-PR-PID-119	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
136	AREA UTILITY	DO-051-AA-4"	TMBP-01-PR-PID-119	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
137	AREA UTILITY	DO-051-AA-4"	TMBP-01-PR-PID-119	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
138	AREA UTILITY	DO-051-AA-4"	TMBP-01-PR-PID-119	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
139	AREA UTILITY	DO-051-AA-4"	TMBP-01-PR-PID-119	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	24	EA	
140	AREA UTILITY	DO-051-AA-4"	TMBP-01-PR-PID-119	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	3	EA	
141	AREA UTILITY	DO-052-AA-4"	TMBP-01-PR-PID-119	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	4,23	M	
142	AREA UTILITY	DO-052-AA-4"	TMBP-01-PR-PID-119	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
143	AREA UTILITY	DO-052-AA-4"	TMBP-01-PR-PID-119	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
144	AREA UTILITY	DO-052-AA-4"	TMBP-01-PR-PID-119	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
145	AREA UTILITY	DO-052-AA-4"	TMBP-01-PR-PID-119	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
146	AREA UTILITY	DO-052-AA-4"	TMBP-01-PR-PID-119	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
147	AREA UTILITY	DO-052-AA-4"	TMBP-01-PR-PID-119	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
148	AREA UTILITY	DO-058-AA-4	TMBP-01-PR-PID-120	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	16,859	M	
149	AREA UTILITY	DO-058-AA-4	TMBP-01-PR-PID-120	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
150	AREA UTILITY	DO-058-AA-4	TMBP-01-PR-PID-120	CONCENTRIC REDUCER	AA	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	3	EA	
151	AREA UTILITY	DO-058-AA-4	TMBP-01-PR-PID-120	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
152	AREA UTILITY	DO-059-AA-6"	TMBP-01-PR-PID-117	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	194,085	M	
153	AREA UTILITY	DO-059-AA-6"	TMBP-01-PR-PID-117	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
154	AREA UTILITY	DO-059-AA-6"	TMBP-01-PR-PID-117	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	4	EA	
155	AREA UTILITY	DO-059-AA-6"	TMBP-01-PR-PID-117	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL 150, RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
156	AREA UTILITY	DO-059-AA-6"	TMBP-01-PR-PID-117	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
157	AREA UTILITY	DO-059-AA-6"	TMBP-01-PR-PID-117	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	2	EA	
158	AREA UTILITY	DO-059-AA-6"	TMBP-01-PR-PID-117	REDUCING TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	2	EA	
159	AREA UTILITY	DO-059-AA-6"	TMBP-01-PR-PID-117	REDUCING TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	1	EA	
160	AREA UTILITY	DO-059-AA-6"	TMBP-01-PR-PID-117	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	24	EA	
161	AREA UTILITY	DO-059-AA-6"	TMBP-01-PR-PID-117	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	3	EA	
162	AREA PRODUKSI	DO-060-AA-4"	TMBP-01-PR-PID-104	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	3,834	M	
163	AREA PRODUKSI	DO-060-AA-4"	TMBP-01-PR-PID-104	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
164	AREA PRODUKSI	DO-060-AA-4"	TMBP-01-PR-PID-104	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
165	AREA PRODUKSI	DO-060-AA-4"	TMBP-01-PR-PID-104	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL 150, RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
166	AREA PRODUKSI	DO-060-AA-4"	TMBP-01-PR-PID-104	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
167	AREA PRODUKSI	DO-060-AA-4"	TMBP-01-PR-PID-104	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
168	AREA PRODUKSI	DO-060-AA-4"	TMBP-01-PR-PID-104	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
169	AREA PRODUKSI	DO-061-AA-4"	TMBP-01-PR-PID-105	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
170	AREA PRODUKSI	DO-061-AA-4"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
171	AREA PRODUKSI	DO-061-AA-4"	TMBP-01-PR-PID-105	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
172	AREA PRODUKSI	DO-061-AA-4"	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
173	AREA PRODUKSI	DO-062-AA-4"	TMBP-01-PR-PID-105	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
174	AREA PRODUKSI	DO-062-AA-4"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
175	AREA PRODUKSI	DO-062-AA-4"	TMBP-01-PR-PID-105	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
176	AREA PRODUKSI	DO-062-AA-4"	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
177	AREA PRODUKSI	DO-063-AA-4"	TMBP-01-PR-PID-106	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	0,314	M	
178	AREA PRODUKSI	DO-063-AA-4"	TMBP-01-PR-PID-106	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
179	AREA PRODUKSI	DO-063-AA-4"	TMBP-01-PR-PID-106	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
180	AREA PRODUKSI	DO-063-AA-4"	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
181	AREA PRODUKSI	DO-063-AA-4"	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
182	AREA PRODUKSI	DO-064-AA-4"	TMBP-01-PR-PID-106	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	0,345	M	
183	AREA PRODUKSI	DO-064-AA-4"	TMBP-01-PR-PID-106	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
184	AREA PRODUKSI	DO-064-AA-4"	TMBP-01-PR-PID-106	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
185	AREA PRODUKSI	DO-064-AA-4"	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
186	AREA PRODUKSI	DO-064-AA-4"	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
187	AREA PRODUKSI	DO-077-AA-4"	TMBP-01-PR-PID-105	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	56,287	M	
188	AREA PRODUKSI	DO-077-AA-4"	TMBP-01-PR-PID-105	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	6	EA	
189	AREA PRODUKSI	DO-077-AA-4"	TMBP-01-PR-PID-105	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	3	EA	
190	AREA PRODUKSI	DO-077-AA-4"	TMBP-01-PR-PID-105	CONCENTRIC REDUCER	AA	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	1	EA	
191	AREA PRODUKSI	DO-077-AA-4"	TMBP-01-PR-PID-105	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL 150, RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
192	AREA PRODUKSI	DO-077-AA-4"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
193	AREA PRODUKSI	DO-077-AA-4"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
194	AREA PRODUKSI	DO-077-AA-4"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	5	EA	
195	AREA PRODUKSI	DO-077-AA-4"	TMBP-01-PR-PID-105	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	6	EA	
196	AREA PRODUKSI	DO-077-AA-4"	TMBP-01-PR-PID-105	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	8	EA	
197	AREA PRODUKSI	DO-077-AA-4"	TMBP-01-PR-PID-105	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	48	EA	
198	AREA PRODUKSI	DO-077-AA-4"	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	1	EA	
199	AREA PRODUKSI	DO-077-AA-4"	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	6	EA	
200	AREA UTILITY	DO-089-AA-4"	TMBP-01-PR-PID-119	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	30,614	M	
201	AREA UTILITY	DO-089-AA-4"	TMBP-01-PR-PID-119	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
202	AREA UTILITY	DO-089-AA-4"	TMBP-01-PR-PID-119	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	6	EA	
203	AREA UTILITY	DO-089-AA-4"	TMBP-01-PR-PID-119	CONCENTRIC REDUCER	AA	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	2	EA	
204	AREA UTILITY	DO-089-AA-4"	TMBP-01-PR-PID-119	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL 150, RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
205	AREA UTILITY	DO-089-AA-4"	TMBP-01-PR-PID-119	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
206	AREA UTILITY	DO-089-AA-4"	TMBP-01-PR-PID-119	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
207	AREA UTILITY	DO-089-AA-4"	TMBP-01-PR-PID-119	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
208	AREA UTILITY	DO-090-AA-6"	TMBP-01-PR-PID-119	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	5,352	M	
209	AREA UTILITY	DO-090-AA-6"	TMBP-01-PR-PID-119	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	5	EA	
210	AREA UTILITY	DO-090-AA-6"	TMBP-01-PR-PID-119	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
211	AREA UTILITY	DO-090-AA-6"	TMBP-01-PR-PID-119	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
212	AREA PRODUKSI	DO-093-AA-4"	TMBP-01-PR-PID-103	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	3,706	M	
213	AREA PRODUKSI	DO-093-AA-4"	TMBP-01-PR-PID-103	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
214	AREA PRODUKSI	DO-093-AA-4"	TMBP-01-PR-PID-103	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	3	EA	
215	AREA PRODUKSI	DO-093-AA-4"	TMBP-01-PR-PID-103	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
216	AREA PRODUKSI	DO-093-AA-4"	TMBP-01-PR-PID-103	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	3	EA	
217	AREA PRODUKSI	DO-093-AA-4"	TMBP-01-PR-PID-103	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	24	EA	
218	AREA PRODUKSI	DO-093-AA-4"	TMBP-01-PR-PID-103	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	3	EA	
219	AREA PRODUKSI	DO-094-AA-4"	TMBP-01-PR-PID-105	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	5,455	M	
220	AREA PRODUKSI	DO-094-AA-4"	TMBP-01-PR-PID-105	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	5	EA	
221	AREA PRODUKSI	DO-094-AA-4"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
222	AREA PRODUKSI	DO-097-AA-4"	TMBP-01-PR-PID-104	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	2,457	M	
223	AREA PRODUKSI	DO-097-AA-4"	TMBP-01-PR-PID-104	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
224	AREA PRODUKSI	DO-097-AA-4"	TMBP-01-PR-PID-104	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
225	AREA PRODUKSI	DO-097-AA-4"	TMBP-01-PR-PID-104	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
226	AREA PRODUKSI	DO-097-AA-4"	TMBP-01-PR-PID-104	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	3	EA	
227	AREA PRODUKSI	DO-097-AA-4"	TMBP-01-PR-PID-104	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	24	EA	
228	AREA PRODUKSI	DO-097-AA-4"	TMBP-01-PR-PID-104	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	3	EA	
229	AREA PRODUKSI	DO-099-AA-4"	TMBP-01-PR-PID-105	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	5,495	M	
230	AREA PRODUKSI	DO-099-AA-4"	TMBP-01-PR-PID-105	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	5	EA	
231	AREA PRODUKSI	DO-099-AA-4"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
232	AREA PRODUKSI	DO-101-AA-6"	TMBP-01-PR-PID-106	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	6,026	M	
233	AREA PRODUKSI	DO-101-AA-6"	TMBP-01-PR-PID-106	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	5	EA	
234	AREA PRODUKSI	DO-101-AA-6"	TMBP-01-PR-PID-106	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
235	AREA PRODUKSI	DO-101-AA-6"	TMBP-01-PR-PID-106	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
236	AREA UTILITY	F-001-AA-3"	TMBP-01-PR-PID-115	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	9,62	M	
237	AREA UTILITY	F-001-AA-3"	TMBP-01-PR-PID-115	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
238	AREA UTILITY	F-001-AA-3"	TMBP-01-PR-PID-115	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	3	EA	
239	AREA UTILITY	F-001-AA-3"	TMBP-01-PR-PID-115	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
240	AREA UTILITY	F-001-AA-3"	TMBP-01-PR-PID-115	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
241	AREA UTILITY	F-001-AA-3"	TMBP-01-PR-PID-115	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
242	AREA UTILITY	F-001-AA-3"	TMBP-01-PR-PID-115	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	

DETAIL MTO PIPES, FLANGES AND FITTINGS (AS PER LINE NUMBER)

JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR

NO.	AREA	LINE NUMBER	PID REFERENCE	PART NAME	MATERIAL CLASS	SCHEDULE	RATING	CONNECTION END	DESCRIPTION	SIZE (Inch)	QTY	UNIT	Remarks
243	AREA UTILITY	F-001-AA-3"	TMBP-01-PR-PID-115	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
244	AREA UTILITY	F-002-AA-3"	TMBP-01-PR-PID-115	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	11,491	M	
245	AREA UTILITY	F-002-AA-3"	TMBP-01-PR-PID-115	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
246	AREA UTILITY	F-002-AA-3"	TMBP-01-PR-PID-115	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
247	AREA UTILITY	F-002-AA-3"	TMBP-01-PR-PID-115	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
248	AREA UTILITY	F-002-AA-3"	TMBP-01-PR-PID-115	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x50mm	8	EA	
249	AREA UTILITY	F-002-AA-3"	TMBP-01-PR-PID-115	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
250	AREA UTILITY	F-003-AA-3"	TMBP-01-PR-PID-115	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	10,297	M	
251	AREA UTILITY	F-003-AA-3"	TMBP-01-PR-PID-115	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
252	AREA UTILITY	F-003-AA-3"	TMBP-01-PR-PID-115	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
253	AREA UTILITY	F-003-AA-3"	TMBP-01-PR-PID-115	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
254	AREA UTILITY	F-003-AA-3"	TMBP-01-PR-PID-115	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
255	AREA UTILITY	F-003-AA-3"	TMBP-01-PR-PID-115	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
256	AREA UTILITY	F-003-AA-3"	TMBP-01-PR-PID-115	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x50mm	8	EA	
257	AREA UTILITY	F-003-AA-3"	TMBP-01-PR-PID-115	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
258	AREA UTILITY	F-004-AA-3"	TMBP-01-PR-PID-115	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	11,537	M	
259	AREA UTILITY	F-004-AA-3"	TMBP-01-PR-PID-115	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
260	AREA UTILITY	F-004-AA-3"	TMBP-01-PR-PID-115	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
261	AREA UTILITY	F-004-AA-3"	TMBP-01-PR-PID-115	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
262	AREA UTILITY	F-004-AA-3"	TMBP-01-PR-PID-115	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x50mm	8	EA	
263	AREA UTILITY	F-004-AA-3"	TMBP-01-PR-PID-115	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
264	AREA UTILITY	F-005-AA-3"	TMBP-01-PR-PID-115	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	55,195	M	
265	AREA UTILITY	F-005-AA-3"	TMBP-01-PR-PID-115	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
266	AREA UTILITY	F-005-AA-3"	TMBP-01-PR-PID-115	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
267	AREA UTILITY	F-005-AA-3"	TMBP-01-PR-PID-115	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
268	AREA UTILITY	F-005-AA-3"	TMBP-01-PR-PID-115	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	4	EA	
269	AREA UTILITY	F-005-AA-3"	TMBP-01-PR-PID-115	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
270	AREA PRODUKSI	F-006-AA-4"	TMBP-01-PR-PID-114	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	0,842	M	
271	AREA PRODUKSI	F-006-AA-4"	TMBP-01-PR-PID-114	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
272	AREA PRODUKSI	F-006-AA-4"	TMBP-01-PR-PID-114	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
273	AREA PRODUKSI	F-006-AA-4"	TMBP-01-PR-PID-114	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x50mm	16	EA	
274	AREA PRODUKSI	F-006-AA-4"	TMBP-01-PR-PID-114	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
275	AREA UTILITY	F-007-AA-4"	TMBP-01-PR-PID-115	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	3,188	M	
276	AREA UTILITY	F-007-AA-4"	TMBP-01-PR-PID-115	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
277	AREA UTILITY	F-007-AA-4"	TMBP-01-PR-PID-115	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
278	AREA UTILITY	F-007-AA-4"	TMBP-01-PR-PID-115	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
279	AREA UTILITY	F-007-AA-4"	TMBP-01-PR-PID-115	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
280	AREA UTILITY	F-007-AA-4"	TMBP-01-PR-PID-115	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x50mm	8	EA	
281	AREA UTILITY	F-007-AA-4"	TMBP-01-PR-PID-115	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
282	AREA UTILITY	F-011-AA-6"	TMBP-01-PR-PID-115	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	147,668	M	
283	AREA UTILITY	F-011-AA-6"	TMBP-01-PR-PID-115	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
284	AREA UTILITY	F-011-AA-6"	TMBP-01-PR-PID-115	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
285	AREA UTILITY	F-011-AA-6"	TMBP-01-PR-PID-115	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	2	EA	
286	AREA UTILITY	F-011-AA-6"	TMBP-01-PR-PID-115	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	10	EA	
287	AREA UTILITY	F-011-AA-6"	TMBP-01-PR-PID-115	CONCENTRIC REDUCER	AA	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"x6"	1	EA	
288	AREA UTILITY	F-011-AA-6"	TMBP-01-PR-PID-115	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	1	EA	
289	AREA UTILITY	F-011-AA-6"	TMBP-01-PR-PID-115	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	2	EA	
290	AREA UTILITY	F-011-AA-6"	TMBP-01-PR-PID-115	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
291	AREA UTILITY	F-011-AA-6"	TMBP-01-PR-PID-115	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	8	EA	
292	AREA UTILITY	F-011-AA-6"	TMBP-01-PR-PID-115	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	16	EA	
293	AREA UTILITY	F-011-AA-6"	TMBP-01-PR-PID-115	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	2	EA	
294	AREA UTILITY	F-011-AA-6"	TMBP-01-PR-PID-115	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	2	EA	
295	AREA UTILITY	F-011-AA-6"	TMBP-01-PR-PID-115	SPECTACLE BLIND	AA		150	RF	SPECTACLE BLIND, A515-70, CL 150, RF, ASME B16.48	8"	1	EA	
296	AREA PRODUKSI	F-012-AA-3"	TMBP-01-PR-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	0,837	M	
297	AREA PRODUKSI	F-012-AA-3"	TMBP-01-PR-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
298	AREA PRODUKSI	F-012-AA-3"	TMBP-01-PR-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
299	AREA PRODUKSI	F-012-AA-3"	TMBP-01-PR-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x50mm	8	EA	
300	AREA PRODUKSI	F-012-AA-3"	TMBP-01-PR-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
301	AREA PRODUKSI	F-013-AA-3"	TMBP-01-PR-PID-102	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	0,742	M	
302	AREA PRODUKSI	F-013-AA-3"	TMBP-01-PR-PID-102	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
303	AREA PRODUKSI	F-013-AA-3"	TMBP-01-PR-PID-102	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
304	AREA PRODUKSI	F-013-AA-3"	TMBP-01-PR-PID-102	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x50mm	8	EA	
305	AREA PRODUKSI	F-013-AA-3"	TMBP-01-PR-PID-102	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
306	AREA PRODUKSI	F-014-AA-4"	TMBP-01-PR-PID-114	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	14,406	M	
307	AREA PRODUKSI	F-014-AA-4"	TMBP-01-PR-PID-114	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
308	AREA PRODUKSI	F-014-AA-4"	TMBP-01-PR-PID-114	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	5	EA	
309	AREA PRODUKSI	F-014-AA-4"	TMBP-01-PR-PID-114	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
310	AREA PRODUKSI	F-014-AA-4"	TMBP-01-PR-PID-114	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
311	AREA PRODUKSI	F-014-AA-4"	TMBP-01-PR-PID-114	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x50mm	16	EA	
312	AREA PRODUKSI	F-014-AA-4"	TMBP-01-PR-PID-114	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
313	AREA UTILITY	F-016-AA-6"	TMBP-01-PR-PID-115	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	132,608	M	
314	AREA UTILITY	F-016-AA-6"	TMBP-01-PR-PID-115	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
315	AREA UTILITY	F-016-AA-6"	TMBP-01-PR-PID-115	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
316	AREA UTILITY	F-016-AA-6"	TMBP-01-PR-PID-115	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	4	EA	
317	AREA UTILITY	F-016-AA-6"	TMBP-01-PR-PID-115	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
318	AREA UTILITY	F-016-AA-6"	TMBP-01-PR-PID-115	CONCENTRIC REDUCER	AA	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	10"x6"	1	EA	
319	AREA UTILITY	F-016-AA-6"	TMBP-01-PR-PID-115	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL 150, RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
320	AREA UTILITY	F-016-AA-6"	TMBP-01-PR-PID-115	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	10"	1	EA	
321	AREA UTILITY	F-016-AA-6"	TMBP-01-PR-PID-115	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
322	AREA UTILITY	F-016-AA-6"	TMBP-01-PR-PID-115	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	8	EA	
323	AREA UTILITY	F-016-AA-6"	TMBP-01-PR-PID-115	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	10"	1	EA	
324	AREA UTILITY	F-016-AA-6"	TMBP-01-PR-PID-115	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	1	EA	
325	AREA UTILITY	F-016-AA-6"	TMBP-01-PR-PID-115	SPECTACLE BLIND	AA		150	RF	SPECTACLE BLIND, A515-70, CL 150, RF, ASME B16.48	10"	1	EA	
326	AREA FIRE WATER	FO-050-AA-3"	#N/A	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	20,383	M	
327	AREA FIRE WATER	FO-050-AA-3"	#N/A	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	3	EA	
328	AREA FIRE WATER	FO-051-AA-3"	#N/A	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	0,821	M	
329	AREA FIRE WATER	FO-051-AA-3"	#N/A	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
330	AREA FIRE WATER	FO-051-AA-3"	#N/A	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
331	AREA FIRE WATER	FO-052-AA-3"	#N/A	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	8,781	M	
332	AREA FIRE WATER	FO-052-AA-3"	#N/A	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
333	AREA FIRE WATER	FO-052-AA-3"	#N/A	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
334	AREA FIRE WATER	FO-053-AA-3"	#N/A	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	61,42	M	
335	AREA FIRE WATER	FO-053-AA-3"	#N/A	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	11	EA	
336	AREA FIRE WATER	FO-053-AA-3"	#N/A	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL 150, RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
337	AREA FIRE WATER	FO-053-AA-3"	#N/A	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
338	AREA FIRE WATER	FO-053-AA-3"	#N/A	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
339	AREA FIRE WATER	FO-053-AA-3"	#N/A	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x50mm	8	EA	
340	AREA FIRE WATER	FO-053-AA-3"	#N/A	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
341	AREA FIRE WATER	FW-001-AA-8"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	8"	2,768	M	
342	AREA FIRE WATER	FW-001-AA-8"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	1	EA	
343	AREA FIRE WATER	FW-001-AA-8"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	5	EA	
344	AREA FIRE WATER	FW-001-AA-8"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
345	AREA FIRE WATER	FW-001-AA-8"	TMBP-01-PS-PID-101	REDUCING TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"x6"	1	EA	
346	AREA FIRE WATER	FW-001-AA-8"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	32	EA	
347	AREA FIRE WATER	FW-001-AA-8"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	4	EA	
348	AREA FIRE WATER	FW-002-AA-8"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	8"	2,768	M	
349	AREA FIRE WATER	FW-002-AA-8"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	1	EA	
350	AREA FIRE WATER	FW-002-AA-8"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	5	EA	
351	AREA FIRE WATER	FW-002-AA-8"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
352	AREA FIRE WATER	FW-002-AA-8"	TMBP-01-PS-PID-101	REDUCING TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"x6"	1	EA	
353	AREA FIRE WATER	FW-002-AA-8"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	32	EA	
354	AREA FIRE WATER	FW-002-AA-8"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	4	EA	
355	AREA FIRE WATER	FW-003-AA-3"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	9,863	M	
356	AREA FIRE WATER	FW-003-AA-3"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
357	AREA FIRE WATER	FW-003-AA-3"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	8	EA	
358	AREA FIRE WATER	FW-003-AA-3"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x50mm	24	EA	
359	AREA FIRE WATER	FW-003-AA-3"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	6	EA	
360	AREA FIRE WATER	FW-004-AA-8"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	8"	17,6	M	
361	AREA FIRE WATER	FW-004-AA-8"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	4	EA	
362	AREA FIRE WATER	FW-004-AA-8"	TMBP-01-PS-PID-101	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL 150, RF, ASME B16.5, SMOOTH FINISHED	8"	1	EA	
363	AREA FIRE WATER	FW-004-AA-8"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	1	EA	

DETAIL MTO PIPES, FLANGES AND FITTINGS (AS PER LINE NUMBER)

JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR

NO.	AREA	LINE NUMBER	PID REFERENCE	PART NAME	MATERIAL CLASS	SCHEDULE	RATING	CONNECTION END	DESCRIPTION	SIZE (Inch)	QTY	UNIT	Remarks
364	AREA FIRE WATER	FW-004-AA-8"	TMBP-01-PS-PID-101	WELDOLET	AA	40		BW	WELDOLET, A105, SCH40, BW, MSS SP-97	8"x3"	1	EA	
365	AREA FIRE WATER	FW-004-AA-8"	TMBP-01-PS-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	3	EA	
366	AREA FIRE WATER	FW-004-AA-8"	TMBP-01-PS-PID-101	REDUCING TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"x6"	1	EA	
367	AREA FIRE WATER	FW-004-AA-8"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	8	EA	
368	AREA FIRE WATER	FW-004-AA-8"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	1	EA	
369	AREA FIRE WATER	FW-005-AA-6"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	0,5	M	
370	AREA FIRE WATER	FW-005-AA-6"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	2	EA	
371	AREA FIRE WATER	FW-005-AA-6"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	8	EA	
372	AREA FIRE WATER	FW-005-AA-6"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	16	EA	
373	AREA FIRE WATER	FW-005-AA-6"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	1	EA	
374	AREA FIRE WATER	FW-005-AA-6"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	2	EA	
375	AREA FIRE WATER	FW-006-AA-6"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	0,5	M	
376	AREA FIRE WATER	FW-006-AA-6"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	2	EA	
377	AREA FIRE WATER	FW-006-AA-6"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	8	EA	
378	AREA FIRE WATER	FW-006-AA-6"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	24	EA	
379	AREA FIRE WATER	FW-006-AA-6"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	1	EA	
380	AREA FIRE WATER	FW-006-AA-6"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	3	EA	
381	AREA FIRE WATER	FW-007-AA-8"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	8"	2,295	M	
382	AREA FIRE WATER	FW-007-AA-8"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	1	EA	
383	AREA FIRE WATER	FW-007-AA-8"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	3	EA	
384	AREA FIRE WATER	FW-007-AA-8"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	16	EA	
385	AREA FIRE WATER	FW-007-AA-8"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	2	EA	
386	AREA FIRE WATER	FW-008-AA-8"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	8"	2,295	M	
387	AREA FIRE WATER	FW-008-AA-8"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	1	EA	
388	AREA FIRE WATER	FW-008-AA-8"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	3	EA	
389	AREA FIRE WATER	FW-008-AA-8"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	16	EA	
390	AREA FIRE WATER	FW-008-AA-8"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	2	EA	
391	AREA FIRE WATER	FW-009-AA-8"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	8"	3,403	M	
392	AREA FIRE WATER	FW-009-AA-8"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	5	EA	
393	AREA FIRE WATER	FW-009-AA-8"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	3	EA	
394	AREA FIRE WATER	FW-009-AA-8"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	16	EA	
395	AREA FIRE WATER	FW-009-AA-8"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	2	EA	
396	AREA FIRE WATER	FW-010-AA-8"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	5,959	M	
397	AREA FIRE WATER	FW-010-AA-8"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	7	EA	
398	AREA FIRE WATER	FW-010-AA-8"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	2	EA	
399	AREA FIRE WATER	FW-010-AA-8"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	16	EA	
400	AREA FIRE WATER	FW-010-AA-8"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	2	EA	
401	AREA FIRE WATER	FW-012-AA-8"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	8"	583,236	M	
402	AREA FIRE WATER	FW-012-AA-8"	TMBP-01-PS-PID-101	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	2	EA	
403	AREA FIRE WATER	FW-012-AA-8"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	25	EA	
404	AREA FIRE WATER	FW-012-AA-8"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	31	EA	
405	AREA FIRE WATER	FW-012-AA-8"	TMBP-01-PS-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	5	EA	
406	AREA FIRE WATER	FW-012-AA-8"	TMBP-01-PS-PID-101	REDUCING TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"x4"	8	EA	
407	AREA FIRE WATER	FW-012-AA-8"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	176	EA	
408	AREA FIRE WATER	FW-012-AA-8"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	22	EA	
409	AREA FIRE WATER	FW-013-AA-8"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	8"	219,977	M	
410	AREA FIRE WATER	FW-013-AA-8"	TMBP-01-PS-PID-101	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	2	EA	
411	AREA FIRE WATER	FW-013-AA-8"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	5	EA	
412	AREA FIRE WATER	FW-013-AA-8"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	7	EA	
413	AREA FIRE WATER	FW-013-AA-8"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	2	EA	
414	AREA FIRE WATER	FW-013-AA-8"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	3	EA	
415	AREA FIRE WATER	FW-013-AA-8"	TMBP-01-PS-PID-101	REDUCING TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"x4"	2	EA	
416	AREA FIRE WATER	FW-013-AA-8"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	16	EA	
417	AREA FIRE WATER	FW-013-AA-8"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	24	EA	
418	AREA FIRE WATER	FW-013-AA-8"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	2	EA	
419	AREA FIRE WATER	FW-013-AA-8"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	3	EA	
420	AREA FIRE WATER	FW-014-AA-8"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	8"	115,955	M	
421	AREA FIRE WATER	FW-014-AA-8"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	4	EA	
422	AREA FIRE WATER	FW-014-AA-8"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	8	EA	
423	AREA FIRE WATER	FW-014-AA-8"	TMBP-01-PS-PID-101	ECCENTRIC REDUCER	AA	40		BW	ECC. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"x6"	1	EA	
424	AREA FIRE WATER	FW-014-AA-8"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	4	EA	
425	AREA FIRE WATER	FW-014-AA-8"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	8	EA	
426	AREA FIRE WATER	FW-014-AA-8"	TMBP-01-PS-PID-101	WELDOLET	AA	40		BW	WELDOLET, A105, SCH40, BW, MSS SP-97	8"x3"	9	EA	
427	AREA FIRE WATER	FW-014-AA-8"	TMBP-01-PS-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	1	EA	
428	AREA FIRE WATER	FW-014-AA-8"	TMBP-01-PS-PID-101	REDUCING TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"x4"	3	EA	
429	AREA FIRE WATER	FW-014-AA-8"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	24	EA	
430	AREA FIRE WATER	FW-014-AA-8"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	48	EA	
431	AREA FIRE WATER	FW-014-AA-8"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	3	EA	
432	AREA FIRE WATER	FW-014-AA-8"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	6	EA	
433	AREA FIRE WATER	FW-015-AA-6"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	31,036	M	
434	AREA FIRE WATER	FW-015-AA-6"	TMBP-01-PS-PID-101	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
435	AREA FIRE WATER	FW-015-AA-6"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
436	AREA FIRE WATER	FW-015-AA-6"	TMBP-01-PS-PID-101	REDUCING TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	1	EA	
437	AREA FIRE WATER	FW-015-AA-6"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	8	EA	
438	AREA FIRE WATER	FW-015-AA-6"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	1	EA	
439	AREA FIRE WATER	FW-017-AA-4"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	0,221	M	
440	AREA FIRE WATER	FW-017-AA-4"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	4,846	M	
441	AREA FIRE WATER	FW-017-AA-4"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
442	AREA FIRE WATER	FW-017-AA-4"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	4	EA	
443	AREA FIRE WATER	FW-017-AA-4"	TMBP-01-PS-PID-101	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
444	AREA FIRE WATER	FW-017-AA-4"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
445	AREA FIRE WATER	FW-017-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	16	EA	
446	AREA FIRE WATER	FW-017-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
447	AREA FIRE WATER	FW-018-AA-4"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	1,378	M	
448	AREA FIRE WATER	FW-018-AA-4"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
449	AREA FIRE WATER	FW-018-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	8	EA	
450	AREA FIRE WATER	FW-018-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	8	EA	
451	AREA FIRE WATER	FW-018-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
452	AREA FIRE WATER	FW-018-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
453	AREA FIRE WATER	FW-019-AA-4"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	18,533	M	
454	AREA FIRE WATER	FW-019-AA-4"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
455	AREA FIRE WATER	FW-019-AA-4"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
456	AREA FIRE WATER	FW-019-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	8	EA	
457	AREA FIRE WATER	FW-019-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	8	EA	
458	AREA FIRE WATER	FW-019-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
459	AREA FIRE WATER	FW-019-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
460	AREA FIRE WATER	FW-020-AA-4"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	2,928	M	
461	AREA FIRE WATER	FW-020-AA-4"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
462	AREA FIRE WATER	FW-020-AA-4"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
463	AREA FIRE WATER	FW-020-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	16	EA	
464	AREA FIRE WATER	FW-020-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
465	AREA FIRE WATER	FW-021-AA-4"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	1,378	M	
466	AREA FIRE WATER	FW-021-AA-4"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
467	AREA FIRE WATER	FW-021-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	8	EA	
468	AREA FIRE WATER	FW-021-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	8	EA	
469	AREA FIRE WATER	FW-021-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
470	AREA FIRE WATER	FW-021-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
471	AREA FIRE WATER	FW-022-AA-4"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	1,378	M	
472	AREA FIRE WATER	FW-022-AA-4"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
473	AREA FIRE WATER	FW-022-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B11.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	16	EA	
474	AREA FIRE WATER	FW-022-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
475	AREA FIRE WATER	FW-023-AA-4"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	1,578	M	
476	AREA FIRE WATER	FW-023-AA-4"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
477	AREA FIRE WATER	FW-023-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.

DETAIL MTO PIPES, FLANGES AND FITTINGS (AS PER LINE NUMBER)

JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR

NO.	AREA	LINE NUMBER	PID REFERENCE	PART NAME	MATERIAL CLASS	SCHEDULE	RATING	CONNECTION END	DESCRIPTION	SIZE (Inch)	QTY	UNIT	Remarks
485	AREA FIRE WATER	FW-024-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
486	AREA FIRE WATER	FW-025-AA-4"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	13,073	M	
487	AREA FIRE WATER	FW-025-AA-4"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
488	AREA FIRE WATER	FW-025-AA-4"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
489	AREA FIRE WATER	FW-025-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
490	AREA FIRE WATER	FW-025-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
491	AREA FIRE WATER	FW-025-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
492	AREA FIRE WATER	FW-025-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
493	AREA FIRE WATER	FW-026-AA-4"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	6,269	M	
494	AREA FIRE WATER	FW-026-AA-4"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
495	AREA FIRE WATER	FW-026-AA-4"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
496	AREA FIRE WATER	FW-026-AA-4"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
497	AREA FIRE WATER	FW-026-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
498	AREA FIRE WATER	FW-026-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
499	AREA FIRE WATER	FW-026-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
500	AREA FIRE WATER	FW-026-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
501	AREA FIRE WATER	FW-027-AA-4"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	0,725	M	
502	AREA FIRE WATER	FW-027-AA-4"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
503	AREA FIRE WATER	FW-027-AA-4"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
504	AREA FIRE WATER	FW-027-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
505	AREA FIRE WATER	FW-027-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
506	AREA FIRE WATER	FW-028-AA-4"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	3,872	M	
507	AREA FIRE WATER	FW-028-AA-4"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
508	AREA FIRE WATER	FW-028-AA-4"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
509	AREA FIRE WATER	FW-028-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
510	AREA FIRE WATER	FW-028-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
511	AREA FIRE WATER	FW-028-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
512	AREA FIRE WATER	FW-028-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
513	AREA FIRE WATER	FW-029-AA-2"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	17,091	M	
514	AREA FIRE WATER	FW-029-AA-2"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	6	EA	
515	AREA FIRE WATER	FW-029-AA-2"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	5	EA	
516	AREA FIRE WATER	FW-029-AA-2"	TMBP-01-PS-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
517	AREA FIRE WATER	FW-029-AA-2"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
518	AREA FIRE WATER	FW-029-AA-2"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
519	AREA FIRE WATER	FW-030-AA-2"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	19,942	M	
520	AREA FIRE WATER	FW-030-AA-2"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	6	EA	
521	AREA FIRE WATER	FW-030-AA-2"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	4	EA	
522	AREA FIRE WATER	FW-030-AA-2"	TMBP-01-PS-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
523	AREA FIRE WATER	FW-030-AA-2"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
524	AREA FIRE WATER	FW-030-AA-2"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
525	AREA FIRE WATER	FW-031-AA-2"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	11,134	M	
526	AREA FIRE WATER	FW-031-AA-2"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	4	EA	
527	AREA FIRE WATER	FW-031-AA-2"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	4	EA	
528	AREA FIRE WATER	FW-031-AA-2"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	4	EA	
529	AREA FIRE WATER	FW-031-AA-2"	TMBP-01-PS-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
530	AREA FIRE WATER	FW-031-AA-2"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
531	AREA FIRE WATER	FW-031-AA-2"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
532	AREA FIRE WATER	FW-032-AA-2"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	11,134	M	
533	AREA FIRE WATER	FW-032-AA-2"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	4	EA	
534	AREA FIRE WATER	FW-032-AA-2"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	4	EA	
535	AREA FIRE WATER	FW-032-AA-2"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	4	EA	
536	AREA FIRE WATER	FW-032-AA-2"	TMBP-01-PS-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
537	AREA FIRE WATER	FW-032-AA-2"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
538	AREA FIRE WATER	FW-032-AA-2"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
539	AREA FIRE WATER	FW-033-AA-3"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	2,04	M	
540	AREA FIRE WATER	FW-033-AA-3"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
541	AREA FIRE WATER	FW-033-AA-3"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	8	EA	
542	AREA FIRE WATER	FW-033-AA-3"	TMBP-01-PS-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
543	AREA FIRE WATER	FW-033-AA-3"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	32	EA	
544	AREA FIRE WATER	FW-033-AA-3"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	8	EA	
545	AREA FIRE WATER	FW-036-AA-3"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	2,04	M	
546	AREA FIRE WATER	FW-036-AA-3"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
547	AREA FIRE WATER	FW-036-AA-3"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	8	EA	
548	AREA FIRE WATER	FW-036-AA-3"	TMBP-01-PS-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
549	AREA FIRE WATER	FW-036-AA-3"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	32	EA	
550	AREA FIRE WATER	FW-036-AA-3"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	8	EA	
551	AREA FIRE WATER	FW-039-AA-4"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	3,271	M	
552	AREA FIRE WATER	FW-039-AA-4"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
553	AREA FIRE WATER	FW-039-AA-4"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
554	AREA FIRE WATER	FW-039-AA-4"	TMBP-01-PS-PID-101	CONCENTRIC REDUCER	AA	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	2	EA	
555	AREA FIRE WATER	FW-039-AA-4"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	6	EA	
556	AREA FIRE WATER	FW-039-AA-4"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
557	AREA FIRE WATER	FW-039-AA-4"	TMBP-01-PS-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
558	AREA FIRE WATER	FW-039-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	48	EA	
559	AREA FIRE WATER	FW-039-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
560	AREA FIRE WATER	FW-039-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	6	EA	
561	AREA FIRE WATER	FW-039-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
562	AREA FIRE WATER	FW-040-AA-4"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	24,969	M	
563	AREA FIRE WATER	FW-040-AA-4"	TMBP-01-PS-PID-101	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
564	AREA FIRE WATER	FW-040-AA-4"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	3	EA	
565	AREA FIRE WATER	FW-041-AA-3"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	4	EA	
566	AREA FIRE WATER	FW-041-AA-3"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	1	EA	
567	AREA FIRE WATER	FW-041-AA-4"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	16,783	M	
568	AREA FIRE WATER	FW-041-AA-4"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	44,47	M	
569	AREA FIRE WATER	FW-041-AA-4"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	6	EA	
570	AREA FIRE WATER	FW-041-AA-4"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
571	AREA FIRE WATER	FW-041-AA-4"	TMBP-01-PS-PID-101	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
572	AREA FIRE WATER	FW-041-AA-4"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
573	AREA FIRE WATER	FW-041-AA-4"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
574	AREA FIRE WATER	FW-041-AA-4"	TMBP-01-PS-PID-101	REDUCNG TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	5	EA	
575	AREA FIRE WATER	FW-041-AA-4"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
576	AREA FIRE WATER	FW-041-AA-4"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
577	AREA FIRE WATER	FW-042-AA-3"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	4,463	M	
578	AREA FIRE WATER	FW-042-AA-3"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	4	EA	
579	AREA FIRE WATER	FW-042-AA-3"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	8	EA	
580	AREA FIRE WATER	FW-042-AA-3"	TMBP-01-PS-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
581	AREA FIRE WATER	FW-042-AA-3"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	32	EA	
582	AREA FIRE WATER	FW-042-AA-3"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	8	EA	
583	AREA FIRE WATER	FW-044-AA-3"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	4,463	M	
584	AREA FIRE WATER	FW-044-AA-3"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	4	EA	
585	AREA FIRE WATER	FW-044-AA-3"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	8	EA	
586	AREA FIRE WATER	FW-044-AA-3"	TMBP-01-PS-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
587	AREA FIRE WATER	FW-044-AA-3"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	32	EA	
588	AREA FIRE WATER	FW-044-AA-3"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	8	EA	
589	AREA FIRE WATER	FW-046-AA-2"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	8,926	M	
590	AREA FIRE WATER	FW-046-AA-2"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	8	EA	
591	AREA FIRE WATER	FW-046-AA-2"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	16	EA	
592	AREA FIRE WATER	FW-046-AA-2"	TMBP-01-PS-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	4	EA	
593	AREA FIRE WATER	FW-046-AA-2"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	64	EA	
594	AREA FIRE WATER	FW-046-AA-2"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	16	EA	
595	AREA FIRE WATER	FW-057-AA-2"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	3,886	M	
596	AREA FIRE WATER	FW-057-AA-2"	TMBP-01-PS-PID-101	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	6	EA	
597	AREA FIRE WATER	FW-057-AA-2"	TMBP-01-PS-PID-101	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
598	AREA FIRE WATER	FW-057-AA-2"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
599	AREA FIRE WATER	FW-057-AA-2"	TMBP-01-PS-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
600	AREA FIRE WATER	FW-057-AA-2"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2/A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
601	AREA FIRE WATER	FW-057-AA-2"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
602	AREA FIRE WATER

DETAIL MTO PIPES, FLANGES AND FITTINGS (AS PER LINE NUMBER)

JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR

NO.	AREA	LINE NUMBER	PID REFERENCE	PART NAME	MATERIAL CLASS	SCHEDULE	RATING	CONNECTION END	DESCRIPTION	SIZE (Inch)	QTY	UNIT	Remarks
606	AREA FIRE WATER	FW-058-AA-3"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	32	EA	
607	AREA FIRE WATER	FW-058-AA-3"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	8	EA	
608	AREA FIRE WATER	FW-060-AA-3"	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	4,463	M	
609	AREA FIRE WATER	FW-060-AA-3"	TMBP-01-PS-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	4	EA	
610	AREA FIRE WATER	FW-060-AA-3"	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	8	EA	
611	AREA FIRE WATER	FW-060-AA-3"	TMBP-01-PS-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
612	AREA FIRE WATER	FW-060-AA-3"	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	32	EA	
613	AREA FIRE WATER	FW-060-AA-3"	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	8	EA	
614	AREA FIRE WATER	FW-064-AA-4"?	TMBP-01-PS-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	1,276	M	
615	AREA FIRE WATER	FW-064-AA-4"?	TMBP-01-PS-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
616	AREA FIRE WATER	FW-064-AA-4"?	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
617	AREA FIRE WATER	FW-064-AA-4"?	TMBP-01-PS-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
618	AREA FIRE WATER	FW-064-AA-4"?	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
619	AREA FIRE WATER	FW-064-AA-4"?	TMBP-01-PS-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
620	AREA PRODUKSI	F-XXX	#N/A	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	258,944	M	
621	AREA PRODUKSI	F-XXX	#N/A	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	6	EA	
622	AREA PRODUKSI	F-XXX	#N/A	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
623	AREA PRODUKSI	F-XXX	#N/A	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	2	EA	
624	AREA PRODUKSI	F-XXX	#N/A	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	8	EA	
625	AREA PRODUKSI	F-XXX	#N/A	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	1	EA	
626	AREA UTILITY	GF-001-AA-4"	TMBP-01-PR-PID-118	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	32,749	M	
627	AREA UTILITY	GF-001-AA-4"	TMBP-01-PR-PID-118	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
628	AREA UTILITY	GF-001-AA-4"	TMBP-01-PR-PID-118	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	3	EA	
629	AREA UTILITY	GF-001-AA-4"	TMBP-01-PR-PID-118	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
630	AREA UTILITY	GF-001-AA-4"	TMBP-01-PR-PID-118	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
631	AREA UTILITY	GF-001-AA-4"	TMBP-01-PR-PID-118	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	4	EA	
632	AREA UTILITY	GF-001-AA-4"	TMBP-01-PR-PID-118	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	13	EA	
633	AREA UTILITY	GF-001-AA-4"	TMBP-01-PR-PID-118	EQUAL TEE	AC	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9, GALV.	4"	1	EA	
634	AREA UTILITY	GF-001-AA-4"	TMBP-01-PR-PID-118	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	5	EA	
635	AREA UTILITY	GF-001-AA-4"	TMBP-01-PR-PID-118	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	120	EA	
636	AREA UTILITY	GF-001-AA-4"	TMBP-01-PR-PID-118	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	15	EA	
637	AREA UTILITY	GF-002-AA-4"	TMBP-01-PR-PID-118	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	6,122	M	
638	AREA UTILITY	GF-002-AA-4"	TMBP-01-PR-PID-118	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
639	AREA UTILITY	GF-002-AA-4"	TMBP-01-PR-PID-118	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	3	EA	
640	AREA UTILITY	GF-002-AA-4"	TMBP-01-PR-PID-118	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
641	AREA UTILITY	GF-002-AA-4"	TMBP-01-PR-PID-118	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	24	EA	
642	AREA UTILITY	GF-002-AA-4"	TMBP-01-PR-PID-118	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	3	EA	
643	AREA UTILITY	GF-003-AA-3"	TMBP-01-PR-PID-118	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	64,165	M	
644	AREA UTILITY	GF-003-AA-3"	TMBP-01-PR-PID-118	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	7	EA	
645	AREA UTILITY	GF-003-AA-3"	TMBP-01-PR-PID-118	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
646	AREA UTILITY	GF-003-AA-3"	TMBP-01-PR-PID-118	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
647	AREA UTILITY	GF-003-AA-3"	TMBP-01-PR-PID-118	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	7	EA	
648	AREA UTILITY	GF-003-AA-3"	TMBP-01-PR-PID-118	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
649	AREA UTILITY	GF-003-AA-3"	TMBP-01-PR-PID-118	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	20	EA	
650	AREA UTILITY	GF-003-AA-3"	TMBP-01-PR-PID-118	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	5	EA	
651	AREA UTILITY	GF-005-AA-3"	TMBP-01-PR-PID-119	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	2,235	M	
652	AREA UTILITY	GF-005-AA-3"	TMBP-01-PR-PID-119	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
653	AREA UTILITY	GF-005-AA-3"	TMBP-01-PR-PID-119	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
654	AREA UTILITY	GF-005-AA-3"	TMBP-01-PR-PID-119	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
655	AREA UTILITY	GF-005-AA-3"	TMBP-01-PR-PID-119	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
656	AREA UTILITY	GF-005-AA-3"	TMBP-01-PR-PID-119	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
657	AREA UTILITY	GF-006-AA-3"	TMBP-01-PR-PID-119	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	59,184	M	
658	AREA UTILITY	GF-006-AA-3"	TMBP-01-PR-PID-119	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	5	EA	
659	AREA UTILITY	GF-006-AA-3"	TMBP-01-PR-PID-119	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
660	AREA UTILITY	GF-006-AA-3"	TMBP-01-PR-PID-119	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
661	AREA UTILITY	GF-006-AA-3"	TMBP-01-PR-PID-119	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
662	AREA UTILITY	GF-006-AA-3"	TMBP-01-PR-PID-119	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	5	EA	
663	AREA UTILITY	GF-006-AA-3"	TMBP-01-PR-PID-119	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
664	AREA UTILITY	GF-006-AA-3"	TMBP-01-PR-PID-119	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
665	AREA UTILITY	GF-006-AA-3"	TMBP-01-PR-PID-119	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	4	EA	
666	AREA UTILITY	GF-009-AA-3	TMBP-01-PR-PID-118	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	103,434	M	
667	AREA UTILITY	GF-009-AA-3	TMBP-01-PR-PID-118	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	11	EA	
668	AREA UTILITY	GF-009-AA-3	TMBP-01-PR-PID-118	ECCENTRIC REDUCER	AA	40		BW	ECC. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	1	EA	
669	AREA UTILITY	GF-009-AA-3	TMBP-01-PR-PID-118	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	3	EA	
670	AREA UTILITY	GF-009-AA-3	TMBP-01-PR-PID-118	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
671	AREA UTILITY	GF-009-AA-3	TMBP-01-PR-PID-118	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
672	AREA PRODUKSI	N10-T-005A	TMBP-01-PR-PID-105	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	8"	1	EA	
673	AREA PRODUKSI	N10-T-005A	TMBP-01-PR-PID-105	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	8	EA	
674	AREA PRODUKSI	N10-T-005A	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	1	EA	
675	AREA PRODUKSI	N10-T-005B	TMBP-01-PR-PID-105	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	8"	1	EA	
676	AREA PRODUKSI	N10-T-005B	TMBP-01-PR-PID-105	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	8	EA	
677	AREA PRODUKSI	N10-T-005B	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	1	EA	
678	AREA PRODUKSI	N10-T-006A	TMBP-01-PR-PID-106	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	8"	1	EA	
679	AREA PRODUKSI	N10-T-006A	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	8	EA	
680	AREA PRODUKSI	N10-T-006A	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	1	EA	
681	AREA PRODUKSI	N10-T-006B	TMBP-01-PR-PID-106	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	8"	1	EA	
682	AREA PRODUKSI	N10-T-006B	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	8	EA	
683	AREA PRODUKSI	N10-T-006B	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	1	EA	
684	AREA PRODUKSI	N12-T-003	TMBP-01-PR-PID-101	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
685	AREA PRODUKSI	N12-T-003	TMBP-01-PR-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
686	AREA PRODUKSI	N12-T-003	TMBP-01-PR-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
687	AREA PRODUKSI	N12-T-004	TMBP-01-PR-PID-101	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
688	AREA PRODUKSI	N12-T-004	TMBP-01-PR-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
689	AREA PRODUKSI	N12-T-004	TMBP-01-PR-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
690	AREA PRODUKSI	N12-T-005A	TMBP-01-PR-PID-105	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
691	AREA PRODUKSI	N12-T-005A	TMBP-01-PR-PID-105	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
692	AREA PRODUKSI	N12-T-005A	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
693	AREA PRODUKSI	N12-T-005B	TMBP-01-PR-PID-105	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
694	AREA PRODUKSI	N12-T-005B	TMBP-01-PR-PID-105	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
695	AREA PRODUKSI	N12-T-005B	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
696	AREA PRODUKSI	N12-T-006A	TMBP-01-PR-PID-106	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
697	AREA PRODUKSI	N12-T-006A	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
698	AREA PRODUKSI	N12-T-006A	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
699	AREA PRODUKSI	N12-T-006B	TMBP-01-PR-PID-106	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
700	AREA PRODUKSI	N12-T-006B	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
701	AREA PRODUKSI	N12-T-006B	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
702	AREA PRODUKSI	N15-T-005A	TMBP-01-PR-PID-105	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	16"	1	EA	
703	AREA PRODUKSI	N15-T-005B	TMBP-01-PR-PID-105	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	16"	1	EA	
704	AREA PRODUKSI	N19-V-001	TMBP-01-PR-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	1,118	M	
705	AREA PRODUKSI	N19-V-001	TMBP-01-PR-PID-101	REDUCING TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	3	EA	
706	AREA PRODUKSI	N2-V-001	TMBP-01-PR-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
707	AREA UTILITY	N6-V-006	TMBP-01-PR-PID-115	CONCENTRIC REDUCER	AA	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	1	EA	
708	AREA UTILITY	N6-V-006	TMBP-01-PR-PID-115	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
709	AREA UTILITY	N6-V-006	TMBP-01-PR-PID-115	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
710	AREA UTILITY	N6-V-006	TMBP-01-PR-PID-115	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL. 150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
711	AREA PRODUKSI	N7-T-003	TMBP-01-PR-PID-101	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	16"	1	EA	
712	AREA PRODUKSI	N7-T-004	TMBP-01-PR-PID-101	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	16"	1	EA	
713	AREA PRODUKSI	N7-V-002	TMBP-01-PR-PID-102	CONCENTRIC REDUCER	AA	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	1	EA	
714	AREA PRODUKSI	N7-V-002	TMBP-01-PR-PID-102	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
715	AREA PRODUKSI	N8-T-006A	TMBP-01-PR-PID-106	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	16"	1	EA	
716	AREA PRODUKSI	N8-T-006B	TMBP-01-PR-PID-106	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	16"	1	EA	
717	AREA PRODUKSI	PF-001-AA-6"	TMBP-01-PR-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	8,874	M	
718	AREA PRODUKSI	PF-001-AA-6"	TMBP-01-PR-PID-101	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
719	AREA PRODUKSI	PF-001-AA-6"	TMBP-01-PR-PID-101	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
720	AREA PRODUKSI	PF-001-AA-6"	TMBP-01-PR-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL. 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	2	EA	
721	AREA PRODUKSI	PF-001-AA-6"	TMBP-01-PR-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
722	AREA PRODUKSI	PF-001-AA-6"	TMBP-01-PR-PID-101	REDUCING TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x3"	12	EA	
723	AREA PRODUKSI	PF-001-AA-6"	TMBP-01-PR-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B

DETAIL MTO PIPES, FLANGES AND FITTINGS (AS PER LINE NUMBER)

JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR

NO.	AREA	LINE NUMBER	PID REFERENCE	PART NAME	MATERIAL CLASS	SCHEDULE	RATING	CONNECTION END	DESCRIPTION	SIZE (Inch)	QTY	UNIT	Remarks
727	AREA PRODUKSI	PF-002-AA-3"	TMBP-01-PR-PID-101	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	3"	12	EA	
728	AREA PRODUKSI	PF-002-AA-3"	TMBP-01-PR-PID-101	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	3"	14	EA	
729	AREA PRODUKSI	PF-002-AA-3"	TMBP-01-PR-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	64	EA	
730	AREA PRODUKSI	PF-002-AA-3"	TMBP-01-PR-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	12	EA	
731	AREA PRODUKSI	PF-002-AA-3"	TMBP-01-PR-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	38	EA	
732	AREA PRODUKSI	PF-002-AA-3"	TMBP-01-PR-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	360	EA	
733	AREA PRODUKSI	PF-002-AA-3"	TMBP-01-PR-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	90	EA	
734	AREA PRODUKSI	PF-003-AA-6"	TMBP-01-PR-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	51,779	M	
735	AREA PRODUKSI	PF-003-AA-6"	TMBP-01-PR-PID-101	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	3	EA	
736	AREA PRODUKSI	PF-003-AA-6"	TMBP-01-PR-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	10	EA	
737	AREA PRODUKSI	PF-003-AA-6"	TMBP-01-PR-PID-101	CONCENTRIC REDUCER	AA	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	12"x6"	1	EA	
738	AREA PRODUKSI	PF-003-AA-6"	TMBP-01-PR-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	12"	1	EA	
739	AREA PRODUKSI	PF-003-AA-6"	TMBP-01-PR-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	16	EA	
740	AREA PRODUKSI	PF-003-AA-6"	TMBP-01-PR-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	4	EA	
741	AREA PRODUKSI	PF-003-AA-6"	TMBP-01-PR-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	7/8"x120mm	12	EA	
742	AREA PRODUKSI	PF-003-AA-6"	TMBP-01-PR-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	120	EA	
743	AREA PRODUKSI	PF-003-AA-6"	TMBP-01-PR-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	12"	2	EA	
744	AREA PRODUKSI	PF-003-AA-6"	TMBP-01-PR-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	15	EA	
745	AREA PRODUKSI	PF-003-AA-6"	TMBP-01-PR-PID-101	SPECTACLE BLIND	AA		150	RF	SPECTACLE BLIND, A515-70, CL.150, RF, ASME B16.48	12"	1	EA	
746	AREA PRODUKSI	PF-004-AA-3"	TMBP-01-PR-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	62,373	M	
747	AREA PRODUKSI	PF-004-AA-3"	TMBP-01-PR-PID-101	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	5	EA	
748	AREA PRODUKSI	PF-004-AA-3"	TMBP-01-PR-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	11	EA	
749	AREA PRODUKSI	PF-004-AA-3"	TMBP-01-PR-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	13	EA	
750	AREA PRODUKSI	PF-004-AA-3"	TMBP-01-PR-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
751	AREA PRODUKSI	PF-004-AA-3"	TMBP-01-PR-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	44	EA	
752	AREA PRODUKSI	PF-004-AA-3"	TMBP-01-PR-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	12	EA	
753	AREA PRODUKSI	PF-004-AA-3"	TMBP-01-PR-PID-101	SPECTACLE BLIND	AA		150	RF	SPECTACLE BLIND, A515-70, CL.150, RF, ASME B16.48	3"	1	EA	
754	AREA PRODUKSI	PF-005-AA-6"	TMBP-01-PR-PID-102	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	8,924	M	
755	AREA PRODUKSI	PF-005-AA-6"	TMBP-01-PR-PID-102	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
756	AREA PRODUKSI	PF-005-AA-6"	TMBP-01-PR-PID-102	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
757	AREA PRODUKSI	PF-005-AA-6"	TMBP-01-PR-PID-102	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	2	EA	
758	AREA PRODUKSI	PF-005-AA-6"	TMBP-01-PR-PID-102	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
759	AREA PRODUKSI	PF-005-AA-6"	TMBP-01-PR-PID-102	REDUCNG TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x3"	12	EA	
760	AREA PRODUKSI	PF-005-AA-6"	TMBP-01-PR-PID-102	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	32	EA	
761	AREA PRODUKSI	PF-005-AA-6"	TMBP-01-PR-PID-102	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	4	EA	
762	AREA PRODUKSI	PF-006-AA-3"	TMBP-01-PR-PID-102	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	59,755	M	
763	AREA PRODUKSI	PF-006-AA-3"	TMBP-01-PR-PID-102	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	3"	12	EA	
764	AREA PRODUKSI	PF-006-AA-3"	TMBP-01-PR-PID-102	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	3"	14	EA	
765	AREA PRODUKSI	PF-006-AA-3"	TMBP-01-PR-PID-102	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	62	EA	
766	AREA PRODUKSI	PF-006-AA-3"	TMBP-01-PR-PID-102	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	12	EA	
767	AREA PRODUKSI	PF-006-AA-3"	TMBP-01-PR-PID-102	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	38	EA	
768	AREA PRODUKSI	PF-006-AA-3"	TMBP-01-PR-PID-102	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	348	EA	
769	AREA PRODUKSI	PF-006-AA-3"	TMBP-01-PR-PID-102	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	87	EA	
770	AREA PRODUKSI	PF-007-AA-6"	TMBP-01-PR-PID-102	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	53,721	M	
771	AREA PRODUKSI	PF-007-AA-6"	TMBP-01-PR-PID-102	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	5	EA	
772	AREA PRODUKSI	PF-007-AA-6"	TMBP-01-PR-PID-102	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	10	EA	
773	AREA PRODUKSI	PF-007-AA-6"	TMBP-01-PR-PID-102	CONCENTRIC REDUCER	AA	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	12"x6"	1	EA	
774	AREA PRODUKSI	PF-007-AA-6"	TMBP-01-PR-PID-102	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	12"	1	EA	
775	AREA PRODUKSI	PF-007-AA-6"	TMBP-01-PR-PID-102	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	16	EA	
776	AREA PRODUKSI	PF-007-AA-6"	TMBP-01-PR-PID-102	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	4	EA	
777	AREA PRODUKSI	PF-007-AA-6"	TMBP-01-PR-PID-102	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	7/8"x120mm	12	EA	
778	AREA PRODUKSI	PF-007-AA-6"	TMBP-01-PR-PID-102	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	120	EA	
779	AREA PRODUKSI	PF-007-AA-6"	TMBP-01-PR-PID-102	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	12"	2	EA	
780	AREA PRODUKSI	PF-007-AA-6"	TMBP-01-PR-PID-102	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	15	EA	
781	AREA PRODUKSI	PF-007-AA-6"	TMBP-01-PR-PID-102	SPECTACLE BLIND	AA		150	RF	SPECTACLE BLIND, A515-70, CL.150, RF, ASME B16.48	12"	1	EA	
782	AREA PRODUKSI	PF-008-AA-3"	TMBP-01-PR-PID-102	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	64,76	M	
783	AREA PRODUKSI	PF-008-AA-3"	TMBP-01-PR-PID-102	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	5	EA	
784	AREA PRODUKSI	PF-008-AA-3"	TMBP-01-PR-PID-102	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	11	EA	
785	AREA PRODUKSI	PF-008-AA-3"	TMBP-01-PR-PID-102	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	13	EA	
786	AREA PRODUKSI	PF-008-AA-3"	TMBP-01-PR-PID-102	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
787	AREA PRODUKSI	PF-008-AA-3"	TMBP-01-PR-PID-102	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	48	EA	
788	AREA PRODUKSI	PF-008-AA-3"	TMBP-01-PR-PID-102	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	12	EA	
789	AREA PRODUKSI	PF-008-AA-3"	TMBP-01-PR-PID-102	SPECTACLE BLIND	AA		150	RF	SPECTACLE BLIND, A515-70, CL.150, RF, ASME B16.48	3"	1	EA	
790	AREA PRODUKSI	PF-009-AA-6"	TMBP-01-PR-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	3,289	M	
791	AREA PRODUKSI	PF-009-AA-6"	TMBP-01-PR-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	4	EA	
792	AREA PRODUKSI	PF-009-AA-6"	TMBP-01-PR-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	2	EA	
793	AREA PRODUKSI	PF-009-AA-6"	TMBP-01-PR-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	2	EA	
794	AREA PRODUKSI	PF-009-AA-6"	TMBP-01-PR-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	32	EA	
795	AREA PRODUKSI	PF-009-AA-6"	TMBP-01-PR-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	4	EA	
796	AREA PRODUKSI	PF-010-AA-6"	TMBP-01-PR-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	2,774	M	
797	AREA PRODUKSI	PF-010-AA-6"	TMBP-01-PR-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	4	EA	
798	AREA PRODUKSI	PF-010-AA-6"	TMBP-01-PR-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	2	EA	
799	AREA PRODUKSI	PF-010-AA-6"	TMBP-01-PR-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	2	EA	
800	AREA PRODUKSI	PF-010-AA-6"	TMBP-01-PR-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	32	EA	
801	AREA PRODUKSI	PF-010-AA-6"	TMBP-01-PR-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	4	EA	
802	AREA PRODUKSI	PG-001-AA-3"	TMBP-01-PR-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	24,173	M	
803	AREA PRODUKSI	PG-001-AA-3"	TMBP-01-PR-PID-101	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
804	AREA PRODUKSI	PG-001-AA-3"	TMBP-01-PR-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	8	EA	
805	AREA PRODUKSI	PG-001-AA-3"	TMBP-01-PR-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	10	EA	
806	AREA PRODUKSI	PG-001-AA-3"	TMBP-01-PR-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
807	AREA PRODUKSI	PG-001-AA-3"	TMBP-01-PR-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	44	EA	
808	AREA PRODUKSI	PG-001-AA-3"	TMBP-01-PR-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	11	EA	
809	AREA PRODUKSI	PG-002-AA-2"	TMBP-01-PR-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
810	AREA PRODUKSI	PG-005-AA-3"	TMBP-01-PR-PID-114	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	20,223	M	
811	AREA PRODUKSI	PG-005-AA-3"	TMBP-01-PR-PID-114	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
812	AREA PRODUKSI	PG-005-AA-3"	TMBP-01-PR-PID-114	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	8	EA	
813	AREA PRODUKSI	PG-005-AA-3"	TMBP-01-PR-PID-114	CONCENTRIC REDUCER	AA	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	1	EA	
814	AREA PRODUKSI	PG-005-AA-3"	TMBP-01-PR-PID-114	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
815	AREA PRODUKSI	PG-005-AA-3"	TMBP-01-PR-PID-114	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	10	EA	
816	AREA PRODUKSI	PG-005-AA-3"	TMBP-01-PR-PID-114	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
817	AREA PRODUKSI	PG-005-AA-3"	TMBP-01-PR-PID-114	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
818	AREA PRODUKSI	PG-005-AA-3"	TMBP-01-PR-PID-114	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	44	EA	
819	AREA PRODUKSI	PG-005-AA-3"	TMBP-01-PR-PID-114	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
820	AREA PRODUKSI	PG-005-AA-3"	TMBP-01-PR-PID-114	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	11	EA	
821	AREA PRODUKSI	PG-005-AA-3"	TMBP-01-PR-PID-114	SPECTACLE BLIND	AA		150	RF	SPECTACLE BLIND, A515-70, CL.150, RF, ASME B16.48	4"	1	EA	
822	AREA PRODUKSI	PG-009-AA-6"	TMBP-01-PR-PID-114	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	0,886	M	
823	AREA PRODUKSI	PG-009-AA-6"	TMBP-01-PR-PID-114	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
824	AREA PRODUKSI	PG-009-AA-6"	TMBP-01-PR-PID-114	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
825	AREA PRODUKSI	PG-009-AA-6"	TMBP-01-PR-PID-114	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	4	EA	
826	AREA PRODUKSI	PG-009-AA-6"	TMBP-01-PR-PID-114	REDUCNG TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x3"	2	EA	
827	AREA PRODUKSI	PG-009-AA-6"	TMBP-01-PR-PID-114	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	32	EA	
828	AREA PRODUKSI	PG-009-AA-6"	TMBP-01-PR-PID-114	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	5	EA	
829	AREA PRODUKSI	PG-009-AA-6"	TMBP-01-PR-PID-114	SPECTACLE BLIND	AA		150	RF	SPECTACLE BLIND, A515-70, CL.150, RF, ASME B16.48	6"	1	EA	
830	AREA PRODUKSI	PG-013-AA-4"	TMBP-01-PR-PID-114	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	6,22	M	
831	AREA PRODUKSI	PG-013-AA-4"	TMBP-01-PR-PID-114	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	5	EA	
832	AREA PRODUKSI	PG-013-AA-4"	TMBP-01-PR-PID-114	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	11	EA	
833	AREA PRODUKSI	PG-013-AA-4"	TMBP-01-PR-PID-114	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
834	AREA PRODUKSI	PG-013-AA-4"	TMBP-01-PR-PID-114	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	88	EA	
835	AREA PRODUKSI	PG-013-AA-4"	TMBP-01-PR-PID-114	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	11	EA	
836	AREA PRODUKSI	PG-016-AA-6"	TMBP-01-PR-PID-114	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	6,455	M	
837	AREA PRODUKSI	PG-016-AA-6"	TMBP-01-PR-PID-114	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	5	EA	
838	AREA PRODUKSI	PG-016-AA-6"	TMBP-01-PR-PID-114	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
839	AREA PRODUKSI	PG-016-AA-6"	TMBP-01-PR-PID-114	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	9	EA	
840	AREA PRODUKSI	PG-016-AA-6"	TMBP-01-PR-PID-114	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
841	AREA PRODUKSI	PG-016-AA-6"	TMBP-01-PR-PID-114	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	2	EA	
842	AREA PRODUKSI	PG-016-AA-6"	TMBP-01-PR-PID-114	REDUCNG TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	1	EA	
843	AREA PRODUKSI	PG-016-AA-6"	TMBP-01-PR-PID-114	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	88	EA	
844	AREA PRODUKSI	PG-016-AA-6"	TMBP-01-PR-PID-114	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	12	EA	
845	AREA PRODUKSI	PG-016-AA-6"	TMBP-01-PR-PID-114	SPECTACLE BLIND	AA		150	RF	SPECTACLE BLIND, A515-70, CL.150, RF, ASME B16.48	6"	1	EA	
846	AREA PRODUKSI	PL-001-AA-3"	TMBP-01-PR-PID-101	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	159,403	M	
847	AREA PRODUKSI	PL-001-AA-3"	TMBP-01-PR-PID-101	45 ELBOW	AA	40		BW	45 ELBOW, A234-WP

DETAIL MTO PIPES, FLANGES AND FITTINGS (AS PER LINE NUMBER)

JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR

NO.	AREA	LINE NUMBER	PID REFERENCE	PART NAME	MATERIAL CLASS	SCHEDULE	RATING	CONNECTION END	DESCRIPTION	SIZE (Inch)	QTY	UNIT	Remarks
848	AREA PRODUKSI	PL-001-AA-3"	TMBP-01-PR-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	8	EA	
849	AREA PRODUKSI	PL-001-AA-3"	TMBP-01-PR-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
850	AREA PRODUKSI	PL-001-AA-3"	TMBP-01-PR-PID-101	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	12	EA	
851	AREA PRODUKSI	PL-001-AA-3"	TMBP-01-PR-PID-101	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
852	AREA PRODUKSI	PL-001-AA-3"	TMBP-01-PR-PID-101	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
853	AREA PRODUKSI	PL-001-AA-3"	TMBP-01-PR-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	24	EA	
854	AREA PRODUKSI	PL-001-AA-3"	TMBP-01-PR-PID-101	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	4	EA	
855	AREA PRODUKSI	PL-001-AA-3"	TMBP-01-PR-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
856	AREA PRODUKSI	PL-001-AA-3"	TMBP-01-PR-PID-101	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	7	EA	
857	AREA PRODUKSI	PL-001-AA-3"	TMBP-01-PR-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	8	EA	
858	AREA PRODUKSI	PL-001-AA-3"	TMBP-01-PR-PID-101	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	120	EA	
859	AREA PRODUKSI	PL-001-AA-3"	TMBP-01-PR-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
860	AREA PRODUKSI	PL-001-AA-3"	TMBP-01-PR-PID-101	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	32	EA	
861	AREA PRODUKSI	PL-001-AA-3"	TMBP-01-PR-PID-101	SPECTACLE BLIND	AA		150	RF	SPECTACLE BLIND, A515-70, CL.150, RF, ASME B16.48	3"	1	EA	
862	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	0,15	M	
863	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	136,44	M	
864	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
865	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
866	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	4	EA	
867	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	14	EA	
868	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	CONCENTRIC REDUCER	AA	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	1	EA	
869	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	CONCENTRIC REDUCER	AA	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	2	EA	
870	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
871	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
872	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
873	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
874	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	22	EA	
875	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	3	EA	
876	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	3	EA	
877	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
878	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
879	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	6	EA	
880	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	REDUCNG TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	1	EA	
881	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	REDUCNG TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	1	EA	
882	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	8	EA	
883	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	216	EA	
884	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	8	EA	
885	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	16	EA	
886	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	2	EA	
887	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	28	EA	
888	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
889	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	4	EA	
890	AREA PRODUKSI	PL-003-AA-4"	TMBP-01-PR-PID-102	SPECTACLE BLIND	AA		150	RF	SPECTACLE BLIND, A515-70, CL.150, RF, ASME B16.48	6"	1	EA	
891	AREA PRODUKSI	PL-006-AA-6"	TMBP-01-PR-PID-103	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	21,332	M	
892	AREA PRODUKSI	PL-006-AA-6"	TMBP-01-PR-PID-103	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
893	AREA PRODUKSI	PL-006-AA-6"	TMBP-01-PR-PID-103	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	4	EA	
894	AREA PRODUKSI	PL-006-AA-6"	TMBP-01-PR-PID-103	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
895	AREA PRODUKSI	PL-006-AA-6"	TMBP-01-PR-PID-103	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
896	AREA PRODUKSI	PL-006-AA-6"	TMBP-01-PR-PID-103	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	4	EA	
897	AREA PRODUKSI	PL-006-AA-6"	TMBP-01-PR-PID-103	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	6	EA	
898	AREA PRODUKSI	PL-006-AA-6"	TMBP-01-PR-PID-103	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
899	AREA PRODUKSI	PL-006-AA-6"	TMBP-01-PR-PID-103	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	3	EA	
900	AREA PRODUKSI	PL-006-AA-6"	TMBP-01-PR-PID-103	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	96	EA	
901	AREA PRODUKSI	PL-006-AA-6"	TMBP-01-PR-PID-103	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	12	EA	
902	AREA PRODUKSI	PL-008-AA-6"	TMBP-01-PR-PID-103	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	1,857	M	
903	AREA PRODUKSI	PL-008-AA-6"	TMBP-01-PR-PID-103	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	0,774	M	
904	AREA PRODUKSI	PL-008-AA-6"	TMBP-01-PR-PID-103	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
905	AREA PRODUKSI	PL-008-AA-6"	TMBP-01-PR-PID-103	ECCENTRIC REDUCER	AA	40		BW	ECC. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x3"	1	EA	
906	AREA PRODUKSI	PL-008-AA-6"	TMBP-01-PR-PID-103	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
907	AREA PRODUKSI	PL-008-AA-6"	TMBP-01-PR-PID-103	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	4	EA	
908	AREA PRODUKSI	PL-008-AA-6"	TMBP-01-PR-PID-103	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
909	AREA PRODUKSI	PL-008-AA-6"	TMBP-01-PR-PID-103	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	40	EA	
910	AREA PRODUKSI	PL-008-AA-6"	TMBP-01-PR-PID-103	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	4	EA	
911	AREA PRODUKSI	PL-008-AA-6"	TMBP-01-PR-PID-103	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	5	EA	
912	AREA PRODUKSI	PL-008-AA-6"	TMBP-01-PR-PID-103	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	1	EA	
913	AREA PRODUKSI	PL-009-AA-3"	TMBP-01-PR-PID-103	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	2,647	M	
914	AREA PRODUKSI	PL-009-AA-3"	TMBP-01-PR-PID-103	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
915	AREA PRODUKSI	PL-009-AA-3"	TMBP-01-PR-PID-103	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
916	AREA PRODUKSI	PL-009-AA-3"	TMBP-01-PR-PID-103	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	8	EA	
917	AREA PRODUKSI	PL-009-AA-3"	TMBP-01-PR-PID-103	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
918	AREA PRODUKSI	PL-010-AA-3"	TMBP-01-PR-PID-103	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	18,682	M	
919	AREA PRODUKSI	PL-010-AA-3"	TMBP-01-PR-PID-103	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	3	EA	
920	AREA PRODUKSI	PL-010-AA-3"	TMBP-01-PR-PID-103	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	4	EA	
921	AREA PRODUKSI	PL-010-AA-3"	TMBP-01-PR-PID-103	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
922	AREA PRODUKSI	PL-010-AA-3"	TMBP-01-PR-PID-103	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	4	EA	
923	AREA PRODUKSI	PL-010-AA-3"	TMBP-01-PR-PID-103	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
924	AREA PRODUKSI	PL-010-AA-3"	TMBP-01-PR-PID-103	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
925	AREA PRODUKSI	PL-010-AA-3"	TMBP-01-PR-PID-103	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	32	EA	
926	AREA PRODUKSI	PL-010-AA-3"	TMBP-01-PR-PID-103	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	8	EA	
927	AREA PRODUKSI	PL-013-AA-3"	TMBP-01-PR-PID-103	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	57,526	M	
928	AREA PRODUKSI	PL-013-AA-3"	TMBP-01-PR-PID-103	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	12	EA	
929	AREA PRODUKSI	PL-013-AA-3"	TMBP-01-PR-PID-103	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	3	EA	
930	AREA PRODUKSI	PL-013-AA-3"	TMBP-01-PR-PID-103	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
931	AREA PRODUKSI	PL-013-AA-3"	TMBP-01-PR-PID-103	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
932	AREA PRODUKSI	PL-013-AA-3"	TMBP-01-PR-PID-103	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
933	AREA PRODUKSI	PL-013-AA-3"	TMBP-01-PR-PID-103	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	4	EA	
934	AREA PRODUKSI	PL-013-AA-3"	TMBP-01-PR-PID-103	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	1	EA	
935	AREA PRODUKSI	PL-015-AA-6"	TMBP-01-PR-PID-104	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	17,327	M	
936	AREA PRODUKSI	PL-015-AA-6"	TMBP-01-PR-PID-104	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
937	AREA PRODUKSI	PL-015-AA-6"	TMBP-01-PR-PID-104	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	2	EA	
938	AREA PRODUKSI	PL-015-AA-6"	TMBP-01-PR-PID-104	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	2	EA	
939	AREA PRODUKSI	PL-015-AA-6"	TMBP-01-PR-PID-104	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
940	AREA PRODUKSI	PL-015-AA-6"	TMBP-01-PR-PID-104	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
941	AREA PRODUKSI	PL-015-AA-6"	TMBP-01-PR-PID-104	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	5	EA	
942	AREA PRODUKSI	PL-015-AA-6"	TMBP-01-PR-PID-104	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	9	EA	
943	AREA PRODUKSI	PL-015-AA-6"	TMBP-01-PR-PID-104	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
944	AREA PRODUKSI	PL-015-AA-6"	TMBP-01-PR-PID-104	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
945	AREA PRODUKSI	PL-015-AA-6"	TMBP-01-PR-PID-104	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	4	EA	
946	AREA PRODUKSI	PL-015-AA-6"	TMBP-01-PR-PID-104	REDUCNG TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x3"	1	EA	
947	AREA PRODUKSI	PL-015-AA-6"	TMBP-01-PR-PID-104	REDUCNG TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x3"	1	EA	
948	AREA PRODUKSI	PL-015-AA-6"	TMBP-01-PR-PID-104	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	128	EA	
949	AREA PRODUKSI	PL-015-AA-6"	TMBP-01-PR-PID-104	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	8	EA	
950	AREA PRODUKSI	PL-015-AA-6"	TMBP-01-PR-PID-104	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	16	EA	
951	AREA PRODUKSI	PL-015-AA-6"	TMBP-01-PR-PID-104	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
952	AREA PRODUKSI	PL-016-AA-6"	TMBP-01-PR-PID-104	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	1,857	M	
953	AREA PRODUKSI	PL-016-AA-6"	TMBP-01-PR-PID-104	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	0,774	M	
954	AREA PRODUKSI	PL-016-AA-6"	TMBP-01-PR-PID-104	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
955	AREA PRODUKSI	PL-016-AA-6"	TMBP-01-PR-PID-104	ECCENTRIC REDUCER	AA	40		BW	ECC. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x3"	1	EA	
956	AREA PRODUKSI	PL-016-AA-6"	TMBP-01-PR-PID-104	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
957	AREA PRODUKSI	PL-016-AA-6"	TMBP-01-PR-PID-104	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	4	EA	
958	AREA PRODUKSI	PL-016-AA-6"	TMBP-01-PR-PID-104	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
959	AREA PRODUKSI	PL-016-AA-6"	TMBP-01-PR-PID-104	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	40	EA	
960	AREA PRODUKSI	PL-016-AA-6"	TMBP-01-PR-PID-104	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	4	EA	
961	AREA PRODUKSI	PL-016-AA-6"	TMBP-01-PR-PID-104	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	5	EA	
962	AREA PRODUKSI	PL-016-AA-6"	TMBP-01-PR-PID-104	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	1	EA	
963	AREA PRODUKSI	PL-017-AA-6"	TMBP-01-PR-PID-104	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	1,857	M	
964	AREA PRODUKSI	PL-017-AA-6"	TMBP-01-PR-PID-104	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	0,774	M	
965	AREA PRODUKSI	PL-017-AA-6"	TMBP-01-PR-PID-104	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
966	AREA PRODUKSI	PL-017-AA-6"	TMBP-01-PR-PID-104	ECCENTRIC REDUCER	AA	40		BW	ECC. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x3"	1	EA	
967	AREA PRODUKSI	PL-017-AA-6"	TMBP-01-PR-PID-104	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
968	AREA PRODUKSI	PL-017-AA-6"	TMBP-01-PR-PID-104	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	4	EA	

DETAIL MTO PIPES, FLANGES AND FITTINGS (AS PER LINE NUMBER)

JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR

NO.	AREA	LINE NUMBER	PID REFERENCE	PART NAME	MATERIAL CLASS	SCHEDULE	RATING	CONNECTION END	DESCRIPTION	SIZE (Inch)	QTY	UNIT	Remarks
969	AREA PRODUKSI	PL-017-AA-6"	TMBP-01-PR-PID-104	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
970	AREA PRODUKSI	PL-017-AA-6"	TMBP-01-PR-PID-104	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	40	EA	
971	AREA PRODUKSI	PL-017-AA-6"	TMBP-01-PR-PID-104	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	4	EA	
972	AREA PRODUKSI	PL-017-AA-6"	TMBP-01-PR-PID-104	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	5	EA	
973	AREA PRODUKSI	PL-017-AA-6"	TMBP-01-PR-PID-104	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	1	EA	
974	AREA PRODUKSI	PL-018-AA-3"	TMBP-01-PR-PID-104	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	4,37	M	
975	AREA PRODUKSI	PL-018-AA-3"	TMBP-01-PR-PID-104	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
976	AREA PRODUKSI	PL-018-AA-3"	TMBP-01-PR-PID-104	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	3	EA	
977	AREA PRODUKSI	PL-018-AA-3"	TMBP-01-PR-PID-104	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	4	EA	
978	AREA PRODUKSI	PL-018-AA-3"	TMBP-01-PR-PID-104	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
979	AREA PRODUKSI	PL-018-AA-3"	TMBP-01-PR-PID-104	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
980	AREA PRODUKSI	PL-018-AA-3"	TMBP-01-PR-PID-104	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	4	EA	
981	AREA PRODUKSI	PL-019-AA-3"	TMBP-01-PR-PID-104	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	4,541	M	
982	AREA PRODUKSI	PL-019-AA-3"	TMBP-01-PR-PID-104	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	4	EA	
983	AREA PRODUKSI	PL-019-AA-3"	TMBP-01-PR-PID-104	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	4	EA	
984	AREA PRODUKSI	PL-019-AA-3"	TMBP-01-PR-PID-104	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
985	AREA PRODUKSI	PL-019-AA-3"	TMBP-01-PR-PID-104	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	4	EA	
986	AREA PRODUKSI	PL-022-AA-3"	TMBP-01-PR-PID-104	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	70,87	M	
987	AREA PRODUKSI	PL-022-AA-3"	TMBP-01-PR-PID-104	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	15	EA	
988	AREA PRODUKSI	PL-022-AA-3"	TMBP-01-PR-PID-104	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
989	AREA PRODUKSI	PL-022-AA-3"	TMBP-01-PR-PID-104	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
990	AREA PRODUKSI	PL-022-AA-3"	TMBP-01-PR-PID-104	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	5	EA	
991	AREA PRODUKSI	PL-022-AA-3"	TMBP-01-PR-PID-104	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	4	EA	
992	AREA PRODUKSI	PL-022-AA-3"	TMBP-01-PR-PID-104	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	24	EA	
993	AREA PRODUKSI	PL-022-AA-3"	TMBP-01-PR-PID-104	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	6	EA	
994	AREA PRODUKSI	PL-024-AA-4"	TMBP-01-PR-PID-105	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	31,649	M	
995	AREA PRODUKSI	PL-024-AA-4"	TMBP-01-PR-PID-105	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	3	EA	
996	AREA PRODUKSI	PL-024-AA-4"	TMBP-01-PR-PID-105	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
997	AREA PRODUKSI	PL-024-AA-4"	TMBP-01-PR-PID-105	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
998	AREA PRODUKSI	PL-024-AA-4"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
999	AREA PRODUKSI	PL-024-AA-4"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
1000	AREA PRODUKSI	PL-024-AA-4"	TMBP-01-PR-PID-105	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
1001	AREA PRODUKSI	PL-024-AA-4"	TMBP-01-PR-PID-105	REDUCING TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	2	EA	
1002	AREA PRODUKSI	PL-024-AA-4"	TMBP-01-PR-PID-105	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	24	EA	
1003	AREA PRODUKSI	PL-024-AA-4"	TMBP-01-PR-PID-105	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
1004	AREA PRODUKSI	PL-024-AA-4"	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	3	EA	
1005	AREA PRODUKSI	PL-024-AA-4"	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
1006	AREA PRODUKSI	PL-025-AA-4"	TMBP-01-PR-PID-105	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	8,815	M	
1007	AREA PRODUKSI	PL-025-AA-4"	TMBP-01-PR-PID-105	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
1008	AREA PRODUKSI	PL-025-AA-4"	TMBP-01-PR-PID-105	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
1009	AREA PRODUKSI	PL-025-AA-4"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
1010	AREA PRODUKSI	PL-025-AA-4"	TMBP-01-PR-PID-105	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
1011	AREA PRODUKSI	PL-025-AA-4"	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
1012	AREA PRODUKSI	PL-026-AA-4"	TMBP-01-PR-PID-105	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	8,844	M	
1013	AREA PRODUKSI	PL-026-AA-4"	TMBP-01-PR-PID-105	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
1014	AREA PRODUKSI	PL-026-AA-4"	TMBP-01-PR-PID-105	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
1015	AREA PRODUKSI	PL-026-AA-4"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
1016	AREA PRODUKSI	PL-026-AA-4"	TMBP-01-PR-PID-105	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
1017	AREA PRODUKSI	PL-026-AA-4"	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	1	EA	
1018	AREA PRODUKSI	PL-027-AA-10"	TMBP-01-PR-PID-105	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	10"	6,051	M	
1019	AREA PRODUKSI	PL-027-AA-10"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	10"	1	EA	
1020	AREA PRODUKSI	PL-027-AA-10"	TMBP-01-PR-PID-105	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	7/8"x115mm	12	EA	
1021	AREA PRODUKSI	PL-027-AA-10"	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	10"	1	EA	
1022	AREA PRODUKSI	PL-028-AA-10"	TMBP-01-PR-PID-105	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	10"	9,115	M	
1023	AREA PRODUKSI	PL-028-AA-10"	TMBP-01-PR-PID-105	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	10"	1	EA	
1024	AREA PRODUKSI	PL-028-AA-10"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	10"	1	EA	
1025	AREA PRODUKSI	PL-028-AA-10"	TMBP-01-PR-PID-105	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	7/8"x115mm	12	EA	
1026	AREA PRODUKSI	PL-028-AA-10"	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	10"	1	EA	
1027	AREA PRODUKSI	PL-029-AA-10"	TMBP-01-PR-PID-105	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	10"	10,044	M	
1028	AREA PRODUKSI	PL-029-AA-10"	TMBP-01-PR-PID-105	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	10"	2	EA	
1029	AREA PRODUKSI	PL-029-AA-10"	TMBP-01-PR-PID-105	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	10"	2	EA	
1030	AREA PRODUKSI	PL-029-AA-10"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	10"	4	EA	
1031	AREA PRODUKSI	PL-029-AA-10"	TMBP-01-PR-PID-105	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	10"	4	EA	
1032	AREA PRODUKSI	PL-029-AA-10"	TMBP-01-PR-PID-105	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	7/8"x115mm	48	EA	
1033	AREA PRODUKSI	PL-029-AA-10"	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	10"	4	EA	
1034	AREA PRODUKSI	PL-030-AA-3"	TMBP-01-PR-PID-105	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	10,44	M	
1035	AREA PRODUKSI	PL-030-AA-3"	TMBP-01-PR-PID-105	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	4	EA	
1036	AREA PRODUKSI	PL-030-AA-3"	TMBP-01-PR-PID-105	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
1037	AREA PRODUKSI	PL-030-AA-3"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
1038	AREA PRODUKSI	PL-030-AA-3"	TMBP-01-PR-PID-105	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	4	EA	
1039	AREA PRODUKSI	PL-030-AA-3"	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	1	EA	
1040	AREA PRODUKSI	PL-031-AA-6"	TMBP-01-PR-PID-106	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	33,86	M	
1041	AREA PRODUKSI	PL-031-AA-6"	TMBP-01-PR-PID-106	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	6"	2	EA	
1042	AREA PRODUKSI	PL-031-AA-6"	TMBP-01-PR-PID-106	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
1043	AREA PRODUKSI	PL-031-AA-6"	TMBP-01-PR-PID-106	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	2	EA	
1044	AREA PRODUKSI	PL-031-AA-6"	TMBP-01-PR-PID-106	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
1045	AREA PRODUKSI	PL-031-AA-6"	TMBP-01-PR-PID-106	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	2	EA	
1046	AREA PRODUKSI	PL-031-AA-6"	TMBP-01-PR-PID-106	REDUCING TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	2	EA	
1047	AREA PRODUKSI	PL-031-AA-6"	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	24	EA	
1048	AREA PRODUKSI	PL-031-AA-6"	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
1049	AREA PRODUKSI	PL-031-AA-6"	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	3	EA	
1050	AREA PRODUKSI	PL-031-AA-6"	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
1051	AREA PRODUKSI	PL-032-AA-6"	TMBP-01-PR-PID-106	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	4,763	M	
1052	AREA PRODUKSI	PL-032-AA-6"	TMBP-01-PR-PID-106	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
1053	AREA PRODUKSI	PL-032-AA-6"	TMBP-01-PR-PID-106	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
1054	AREA PRODUKSI	PL-032-AA-6"	TMBP-01-PR-PID-106	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
1055	AREA PRODUKSI	PL-032-AA-6"	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	8	EA	
1056	AREA PRODUKSI	PL-032-AA-6"	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	1	EA	
1057	AREA PRODUKSI	PL-033-AA-6"	TMBP-01-PR-PID-106	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	4,019	M	
1058	AREA PRODUKSI	PL-033-AA-6"	TMBP-01-PR-PID-106	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
1059	AREA PRODUKSI	PL-033-AA-6"	TMBP-01-PR-PID-106	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
1060	AREA PRODUKSI	PL-033-AA-6"	TMBP-01-PR-PID-106	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
1061	AREA PRODUKSI	PL-033-AA-6"	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	8	EA	
1062	AREA PRODUKSI	PL-033-AA-6"	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	1	EA	
1063	AREA PRODUKSI	PL-034-AA-10"	TMBP-01-PR-PID-106	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	10"	2,063	M	
1064	AREA PRODUKSI	PL-034-AA-10"	TMBP-01-PR-PID-106	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	10"	1	EA	
1065	AREA PRODUKSI	PL-034-AA-10"	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	7/8"x115mm	24	EA	
1066	AREA PRODUKSI	PL-034-AA-10"	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	10"	2	EA	
1067	AREA PRODUKSI	PL-035-AA-10"	TMBP-01-PR-PID-106	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	10"	1,517	M	
1068	AREA PRODUKSI	PL-035-AA-10"	TMBP-01-PR-PID-106	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	10"	1	EA	
1069	AREA PRODUKSI	PL-035-AA-10"	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	7/8"x115mm	24	EA	
1070	AREA PRODUKSI	PL-035-AA-10"	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	10"	2	EA	
1071	AREA PRODUKSI	PL-036-AA-10"	TMBP-01-PR-PID-106	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	10"	16,775	M	
1072	AREA PRODUKSI	PL-036-AA-10"	TMBP-01-PR-PID-106	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	10"	1	EA	
1073	AREA PRODUKSI	PL-036-AA-10"	TMBP-01-PR-PID-106	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	10"	2	EA	
1074	AREA PRODUKSI	PL-036-AA-10"	TMBP-01-PR-PID-106	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	10"	1	EA	
1075	AREA PRODUKSI	PL-036-AA-10"	TMBP-01-PR-PID-106	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	10"	7	EA	
1076	AREA PRODUKSI	PL-036-AA-10"	TMBP-01-PR-PID-106	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	10"	4	EA	
1077	AREA PRODUKSI	PL-036-AA-10"	TMBP-01-PR-PID-106	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	10"	2	EA	
1078	AREA PRODUKSI	PL-036-AA-10"	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	7/8"x115mm	120	EA	
1079	AREA PRODUKSI	PL-036-AA-10"	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	10"	10	EA	
1080	AREA PRODUKSI	PL-037-AA-3"	TMBP-01-PR-PID-106	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	12,778	M	
1081	AREA PRODUKSI	PL-037-AA-3"	TMBP-01-PR-PID-106	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	4	EA	
1082	AREA PRODUKSI	PL-037-AA-3"	TMBP-01-PR-PID-106	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
1083	AREA PRODUKSI	PL-037-AA-3"	TMBP-01-PR-PID-106	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	4	EA	
1084	AREA PRODUKSI	PL-037-AA-3"	TMBP-01-PR-PID-106	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	1	EA	
1085	AREA UTILITY	PL-040-AA-3"	TMBP-01-PR-PID-115	PIPE	AA	40		B

DETAIL MTO PIPES, FLANGES AND FITTINGS (AS PER LINE NUMBER)

JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR

NO.	AREA	LINE NUMBER	PID REFERENCE	PART NAME	MATERIAL CLASS	SCHEDULE	RATING	CONNECTION END	DESCRIPTION	SIZE (Inch)	QTY	UNIT	Remarks
1090	AREA UTILITY	PL-040-AA-3"	TMBP-01-PR-PID-115	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
1091	AREA UTILITY	PL-040-AA-3"	TMBP-01-PR-PID-115	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	28	EA	
1092	AREA UTILITY	PL-040-AA-3"	TMBP-01-PR-PID-115	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	7	EA	
1093	AREA UTILITY	PL-041-AA-3"	TMBP-01-PR-PID-115	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	1,343	M	
1094	AREA UTILITY	PL-041-AA-3"	TMBP-01-PR-PID-115	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
1095	AREA UTILITY	PL-041-AA-3"	TMBP-01-PR-PID-115	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
1096	AREA UTILITY	PL-041-AA-3"	TMBP-01-PR-PID-115	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
1097	AREA UTILITY	PL-041-AA-3"	TMBP-01-PR-PID-115	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	4	EA	
1098	AREA UTILITY	PL-041-AA-3"	TMBP-01-PR-PID-115	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	20	EA	
1099	AREA UTILITY	PL-041-AA-3"	TMBP-01-PR-PID-115	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	5	EA	
1100	AREA UTILITY	PL-043-AA-3"	TMBP-01-PR-PID-116	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	1,962	M	
1101	AREA UTILITY	PL-043-AA-3"	TMBP-01-PR-PID-116	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
1102	AREA UTILITY	PL-043-AA-3"	TMBP-01-PR-PID-116	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	2	EA	
1103	AREA UTILITY	PL-043-AA-3"	TMBP-01-PR-PID-116	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	4	EA	
1104	AREA UTILITY	PL-043-AA-3"	TMBP-01-PR-PID-116	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	4	EA	
1105	AREA UTILITY	PL-043-AA-3"	TMBP-01-PR-PID-116	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
1106	AREA UTILITY	PL-043-AA-3"	TMBP-01-PR-PID-116	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	24	EA	
1107	AREA UTILITY	PL-043-AA-3"	TMBP-01-PR-PID-116	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	7	EA	
1108	AREA UTILITY	PL-043-AA-3"	TMBP-01-PR-PID-116	SPECTACLE BLIND	AA		150	RF	SPECTACLE BLIND, A515-70, CL.150, RF, ASME B16.48	3"	1	EA	
1109	AREA UTILITY	PL-044-AA-3"	TMBP-01-PR-PID-116	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	1,349	M	
1110	AREA UTILITY	PL-044-AA-3"	TMBP-01-PR-PID-116	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
1111	AREA UTILITY	PL-044-AA-3"	TMBP-01-PR-PID-116	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
1112	AREA UTILITY	PL-044-AA-3"	TMBP-01-PR-PID-116	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
1113	AREA UTILITY	PL-044-AA-3"	TMBP-01-PR-PID-116	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
1114	AREA UTILITY	PL-044-AA-3"	TMBP-01-PR-PID-116	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
1115	AREA UTILITY	PL-044-AA-3"	TMBP-01-PR-PID-116	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	4	EA	
1116	AREA PRODUKSI	PL-046-AA-10"	TMBP-01-PR-PID-107	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	10"	0,705	M	
1117	AREA PRODUKSI	PL-046-AA-10"	TMBP-01-PR-PID-107	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	10"	1	EA	
1118	AREA PRODUKSI	PL-046-AA-10"	TMBP-01-PR-PID-107	FLANGE	AA	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	10"	1	EA	
1119	AREA PRODUKSI	PL-047-AA-10"	TMBP-01-PR-PID-107	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	10"	0,705	M	
1120	AREA PRODUKSI	PL-047-AA-10"	TMBP-01-PR-PID-107	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	10"	1	EA	
1121	AREA PRODUKSI	PL-047-AA-10"	TMBP-01-PR-PID-107	FLANGE	AA	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	10"	1	EA	
1122	AREA PRODUKSI	PL-048-BA-6"	TMBP-01-PR-PID-107	PIPE	BA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	4,83	M	
1123	AREA PRODUKSI	PL-048-BA-6"	TMBP-01-PR-PID-107	90 ELBOW	BA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	4	EA	
1124	AREA PRODUKSI	PL-048-BA-6"	TMBP-01-PR-PID-107	CONCENTRIC REDUCER	BA	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	1	EA	
1125	AREA PRODUKSI	PL-048-BA-6"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	4	EA	
1126	AREA PRODUKSI	PL-048-BA-6"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
1127	AREA PRODUKSI	PL-048-BA-6"	TMBP-01-PR-PID-107	REDUCING TEE	BA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x3"	2	EA	
1128	AREA PRODUKSI	PL-048-BA-6"	TMBP-01-PR-PID-107	BOLT&NUTS	BA		300		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x120mm	48	EA	
1129	AREA PRODUKSI	PL-048-BA-6"	TMBP-01-PR-PID-107	BOLT&NUTS	BA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
1130	AREA PRODUKSI	PL-048-BA-6"	TMBP-01-PR-PID-107	GASKET	BA		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK3mm INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	4	EA	
1131	AREA PRODUKSI	PL-049-BA-6"	TMBP-01-PR-PID-107	PIPE	BA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	4,83	M	
1132	AREA PRODUKSI	PL-049-BA-6"	TMBP-01-PR-PID-107	90 ELBOW	BA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	4	EA	
1133	AREA PRODUKSI	PL-049-BA-6"	TMBP-01-PR-PID-107	CONCENTRIC REDUCER	BA	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	1	EA	
1134	AREA PRODUKSI	PL-049-BA-6"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	4	EA	
1135	AREA PRODUKSI	PL-049-BA-6"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
1136	AREA PRODUKSI	PL-049-BA-6"	TMBP-01-PR-PID-107	REDUCING TEE	BA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	2	EA	
1137	AREA PRODUKSI	PL-049-BA-6"	TMBP-01-PR-PID-107	BOLT&NUTS	BA		300		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x120mm	48	EA	
1138	AREA PRODUKSI	PL-049-BA-6"	TMBP-01-PR-PID-107	BOLT&NUTS	BA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
1139	AREA PRODUKSI	PL-049-BA-6"	TMBP-01-PR-PID-107	GASKET	BA		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK3mm INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	4	EA	
1140	AREA PRODUKSI	PL-050-BA-4"	TMBP-01-PR-PID-107	PIPE	BA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	2,558	M	
1141	AREA PRODUKSI	PL-050-BA-4"	TMBP-01-PR-PID-107	90 ELBOW	BA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
1142	AREA PRODUKSI	PL-050-BA-4"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
1143	AREA PRODUKSI	PL-050-BA-4"	TMBP-01-PR-PID-107	BOLT&NUTS	BA		300		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x115mm	16	EA	
1144	AREA PRODUKSI	PL-050-BA-4"	TMBP-01-PR-PID-107	GASKET	BA		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK3mm INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
1145	AREA PRODUKSI	PL-051-AA-4"	TMBP-01-PR-PID-107	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	30,465	M	
1146	AREA PRODUKSI	PL-051-AA-4"	TMBP-01-PR-PID-107	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	4	EA	
1147	AREA PRODUKSI	PL-051-AA-4"	TMBP-01-PR-PID-107	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	6	EA	
1148	AREA PRODUKSI	PL-051-AA-4"	TMBP-01-PR-PID-107	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
1149	AREA PRODUKSI	PL-051-AA-4"	TMBP-01-PR-PID-107	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	7	EA	
1150	AREA PRODUKSI	PL-051-AA-4"	TMBP-01-PR-PID-107	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
1151	AREA PRODUKSI	PL-051-AA-4"	TMBP-01-PR-PID-107	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	5	EA	
1152	AREA PRODUKSI	PL-051-AA-4"	TMBP-01-PR-PID-107	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	48	EA	
1153	AREA PRODUKSI	PL-051-AA-4"	TMBP-01-PR-PID-107	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	6	EA	
1154	AREA PRODUKSI	PL-052-BA-6"	TMBP-01-PR-PID-107	PIPE	BA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	99,776	M	
1155	AREA PRODUKSI	PL-052-BA-6"	TMBP-01-PR-PID-107	90 ELBOW	BA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	9	EA	
1156	AREA PRODUKSI	PL-052-BA-6"	TMBP-01-PR-PID-107	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
1157	AREA PRODUKSI	PL-052-BA-6"	TMBP-01-PR-PID-107	BLIND FLANGE	BA		300	RF	BLIND FLANGE, A105, CL.300, RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
1158	AREA PRODUKSI	PL-052-BA-6"	TMBP-01-PR-PID-107	BLIND FLANGE	BA		300	RF	BLIND FLANGE, A105, CL.300, RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
1159	AREA PRODUKSI	PL-052-BA-6"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	12	EA	
1160	AREA PRODUKSI	PL-052-BA-6"	TMBP-01-PR-PID-107	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
1161	AREA PRODUKSI	PL-052-BA-6"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	2	EA	
1162	AREA PRODUKSI	PL-052-BA-6"	TMBP-01-PR-PID-107	EQUAL TEE	BA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	4	EA	
1163	AREA PRODUKSI	PL-052-BA-6"	TMBP-01-PR-PID-107	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	1	EA	
1164	AREA PRODUKSI	PL-052-BA-6"	TMBP-01-PR-PID-107	REDUCING TEE	BA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	1	EA	
1165	AREA PRODUKSI	PL-052-BA-6"	TMBP-01-PR-PID-107	BOLT&NUTS	BA		300		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x120mm	144	EA	
1166	AREA PRODUKSI	PL-052-BA-6"	TMBP-01-PR-PID-107	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	8	EA	
1167	AREA PRODUKSI	PL-052-BA-6"	TMBP-01-PR-PID-107	GASKET	BA		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK3mm INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	9	EA	
1168	AREA PRODUKSI	PL-052-BA-6"	TMBP-01-PR-PID-107	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	1	EA	
1169	AREA PRODUKSI	PL-052-BA-6"	TMBP-01-PR-PID-107	GASKET	BA		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK3mm INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	4	EA	
1170	AREA PRODUKSI	PL-053-AA-3"	TMBP-01-PR-PID-105	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	12,219	M	
1171	AREA PRODUKSI	PL-053-AA-3"	TMBP-01-PR-PID-105	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
1172	AREA PRODUKSI	PL-053-AA-3"	TMBP-01-PR-PID-105	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	7	EA	
1173	AREA PRODUKSI	PL-053-AA-3"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
1174	AREA PRODUKSI	PL-053-AA-3"	TMBP-01-PR-PID-105	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
1175	AREA PRODUKSI	PL-053-AA-3"	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
1176	AREA PRODUKSI	PL-054-AA-4"	TMBP-01-PR-PID-107	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	11,006	M	
1177	AREA PRODUKSI	PL-054-AA-4"	TMBP-01-PR-PID-107	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	6	EA	
1178	AREA PRODUKSI	PL-054-AA-4"	TMBP-01-PR-PID-107	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
1179	AREA PRODUKSI	PL-054-AA-4"	TMBP-01-PR-PID-107	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	16	EA	
1180	AREA PRODUKSI	PL-054-AA-4"	TMBP-01-PR-PID-107	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
1181	AREA PRODUKSI	PL-055-BA-3"	TMBP-01-PR-PID-107	PIPE	BA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	0,472	M	
1182	AREA PRODUKSI	PL-055-BA-3"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
1183	AREA PRODUKSI	PL-055-BA-3"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
1184	AREA PRODUKSI	PL-055-BA-3"	TMBP-01-PR-PID-107	EQUAL TEE	BA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
1185	AREA PRODUKSI	PL-055-BA-3"	TMBP-01-PR-PID-107	BOLT&NUTS	BA		300		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	24	EA	
1186	AREA PRODUKSI	PL-055-BA-3"	TMBP-01-PR-PID-107	BOLT&NUTS	AA		300		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	8	EA	
1187	AREA PRODUKSI	PL-055-BA-3"	TMBP-01-PR-PID-107	BOLT&NUTS	BA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
1188	AREA PRODUKSI	PL-055-BA-3"	TMBP-01-PR-PID-107	GASKET	BA		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK3mm INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
1189	AREA PRODUKSI	PL-055-BA-3"	TMBP-01-PR-PID-107	GASKET	BA		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK3mm INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
1190	AREA PRODUKSI	PL-056-BA-3"	TMBP-01-PR-PID-107	PIPE	BA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	0,472	M	
1191	AREA PRODUKSI	PL-056-BA-3"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
1192	AREA PRODUKSI	PL-056-BA-3"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
1193	AREA PRODUKSI	PL-056-BA-3"	TMBP-01-PR-PID-107	EQUAL TEE	BA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	1	EA	
1194	AREA PRODUKSI	PL-056-BA-3"	TMBP-01-PR-PID-107	BOLT&NUTS	BA		300		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	32	EA	
1195	AREA PRODUKSI	PL-056-BA-3"	TMBP-01-PR-PID-107	BOLT&NUTS	AA		300		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	8	EA	
1196	AREA PRODUKSI	PL-056-BA-3"	TMBP-01-PR-PID-107	BOLT&NUTS	BA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x60mm	8	EA	
1197	AREA PRODUKSI	PL-056-BA-3"	TMBP-01-PR-PID-107	GASKET	BA		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK3mm INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	2	EA	
1198	AREA PRODUKSI	PL-056-BA-3"	TMBP-01-PR-PID-107	GASKET	AA		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	3	EA	
1199	AREA PRODUKSI	PL-057-AA-4"	TMBP-01-PR-PID-107	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	1,144	M	
1200	AREA PRODUKSI	PL-057-AA-4"	TMBP-01-PR-PID-107	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
1201	AREA PRODUKSI	PL-057-AA-4"	TMBP-01-PR-PID-107	ECCENTRIC REDUCER	AA	40		BW	ECC. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	1	EA	
1202	AREA PRODUKSI	PL-057-AA-4"	TMBP-01-PR-PID-107	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
1203	AREA PRODUKSI	PL-057-AA-4"	TMBP-01-PR-PID-107	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
1204	AREA PRODUKSI	PL-057-AA-4"	TMBP-01-PR-PID-107	FLANGE	AA	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
1205	AREA PRODUKSI	PL-057-AA-4"	TMBP-01-PR-PID-107	FLANGE	AA	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
1206	AREA PRODUKSI	PL-057-AA-4"	TMBP-01-PR-PID-107	REDUCING TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	1	EA	
1207	AREA PRODUKSI	PL-057-AA-4"	TMBP-01-PR-PID-107	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.

DETAIL MTO PIPES, FLANGES AND FITTINGS (AS PER LINE NUMBER)

JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR

NO.	AREA	LINE NUMBER	PID REFERENCE	PART NAME	MATERIAL CLASS	SCHEDULE	RATING	CONNECTION END	DESCRIPTION	SIZE (Inch)	QTY	UNIT	Remarks
1211	AREA PRODUKSI	PL-058-AA-4"	TMBP-01-PR-PID-107	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	1,144	M	
1212	AREA PRODUKSI	PL-058-AA-4"	TMBP-01-PR-PID-107	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
1213	AREA PRODUKSI	PL-058-AA-4"	TMBP-01-PR-PID-107	ECCENTRIC REDUCER	AA	40		BW	ECC. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	1	EA	
1214	AREA PRODUKSI	PL-058-AA-4"	TMBP-01-PR-PID-107	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
1215	AREA PRODUKSI	PL-058-AA-4"	TMBP-01-PR-PID-107	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
1216	AREA PRODUKSI	PL-058-AA-4"	TMBP-01-PR-PID-107	FLANGE	AA	40	300	WN-RF	FLANGE, A105, CL 300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	1	EA	
1217	AREA PRODUKSI	PL-058-AA-4"	TMBP-01-PR-PID-107	REDUCING TEE	AA	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	1	EA	
1218	AREA PRODUKSI	PL-058-AA-4"	TMBP-01-PR-PID-107	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	16	EA	
1219	AREA PRODUKSI	PL-058-AA-4"	TMBP-01-PR-PID-107	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
1220	AREA PRODUKSI	PL-059-AA-4"	TMBP-01-PR-PID-107	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	29,371	M	
1221	AREA PRODUKSI	PL-059-AA-4"	TMBP-01-PR-PID-107	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
1222	AREA PRODUKSI	PL-059-AA-4"	TMBP-01-PR-PID-107	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	5	EA	
1223	AREA PRODUKSI	PL-059-AA-4"	TMBP-01-PR-PID-107	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	3	EA	
1224	AREA PRODUKSI	PL-059-AA-4"	TMBP-01-PR-PID-107	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
1225	AREA PRODUKSI	PL-059-AA-4"	TMBP-01-PR-PID-107	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
1226	AREA PRODUKSI	PL-059-AA-4"	TMBP-01-PR-PID-107	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	16	EA	
1227	AREA PRODUKSI	PL-059-AA-4"	TMBP-01-PR-PID-107	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
1228	AREA PRODUKSI	PL-060-AA-4"	TMBP-01-PR-PID-107	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	14,061	M	
1229	AREA PRODUKSI	PL-060-AA-4"	TMBP-01-PR-PID-107	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	4	EA	
1230	AREA PRODUKSI	PL-060-AA-4"	TMBP-01-PR-PID-107	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
1231	AREA PRODUKSI	PL-060-AA-4"	TMBP-01-PR-PID-107	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
1232	AREA PRODUKSI	PL-060-AA-4"	TMBP-01-PR-PID-107	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	24	EA	
1233	AREA PRODUKSI	PL-060-AA-4"	TMBP-01-PR-PID-107	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	3	EA	
1234	AREA PRODUKSI	PL-061-AA-4"	TMBP-01-PR-PID-107	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	18,703	M	
1235	AREA PRODUKSI	PL-061-AA-4"	TMBP-01-PR-PID-107	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	3	EA	
1236	AREA PRODUKSI	PL-061-AA-4"	TMBP-01-PR-PID-107	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	5	EA	
1237	AREA PRODUKSI	PL-061-AA-4"	TMBP-01-PR-PID-107	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
1238	AREA PRODUKSI	PL-061-AA-4"	TMBP-01-PR-PID-107	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	24	EA	
1239	AREA PRODUKSI	PL-061-AA-4"	TMBP-01-PR-PID-107	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	3	EA	
1240	AREA PRODUKSI	PL-062-BA-4"	TMBP-01-PR-PID-107	PIPE	BA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	22,067	M	
1241	AREA PRODUKSI	PL-062-BA-4"	TMBP-01-PR-PID-107	90 ELBOW	BA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	6	EA	
1242	AREA PRODUKSI	PL-062-BA-4"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL 300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	4	EA	
1243	AREA PRODUKSI	PL-062-BA-4"	TMBP-01-PR-PID-107	BOLT&NUTS	BA		300		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x115mm	32	EA	
1244	AREA PRODUKSI	PL-062-BA-4"	TMBP-01-PR-PID-107	GASKET	BA		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK3mm INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	4	EA	
1245	AREA PRODUKSI	PL-063-BA-10"	TMBP-01-PR-PID-107	PIPE	BA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	10"	2,049	M	
1246	AREA PRODUKSI	PL-063-BA-10"	TMBP-01-PR-PID-107	PIPE	BA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	8"	0,634	M	
1247	AREA PRODUKSI	PL-063-BA-10"	TMBP-01-PR-PID-107	ECCENTRIC REDUCER	BA	40		BW	ECC. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	10"x8"	1	EA	
1248	AREA PRODUKSI	PL-063-BA-10"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL 300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	10"	2	EA	
1249	AREA PRODUKSI	PL-063-BA-10"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL 300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	10"	1	EA	
1250	AREA PRODUKSI	PL-063-BA-10"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL 300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	1	EA	
1251	AREA PRODUKSI	PL-063-BA-10"	TMBP-01-PR-PID-107	BOLT&NUTS	BA		300		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	1"x160mm	64	EA	
1252	AREA PRODUKSI	PL-063-BA-10"	TMBP-01-PR-PID-107	BOLT&NUTS	BA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	8	EA	
1253	AREA PRODUKSI	PL-063-BA-10"	TMBP-01-PR-PID-107	GASKET	BA		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK3mm INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	10"	2	EA	
1254	AREA PRODUKSI	PL-063-BA-10"	TMBP-01-PR-PID-107	GASKET	BA		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK3mm INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	10"	2	EA	
1255	AREA PRODUKSI	PL-064-BA-10"	TMBP-01-PR-PID-107	PIPE	BA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	10"	2,049	M	
1256	AREA PRODUKSI	PL-064-BA-10"	TMBP-01-PR-PID-107	PIPE	BA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	8"	0,634	M	
1257	AREA PRODUKSI	PL-064-BA-10"	TMBP-01-PR-PID-107	ECCENTRIC REDUCER	BA	40		BW	ECC. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	10"x8"	1	EA	
1258	AREA PRODUKSI	PL-064-BA-10"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL 300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	10"	2	EA	
1259	AREA PRODUKSI	PL-064-BA-10"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL 300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	10"	1	EA	
1260	AREA PRODUKSI	PL-064-BA-10"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL 300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	1	EA	
1261	AREA PRODUKSI	PL-064-BA-10"	TMBP-01-PR-PID-107	BOLT&NUTS	BA		300		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	1"x160mm	64	EA	
1262	AREA PRODUKSI	PL-064-BA-10"	TMBP-01-PR-PID-107	BOLT&NUTS	BA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	8	EA	
1263	AREA PRODUKSI	PL-064-BA-10"	TMBP-01-PR-PID-107	GASKET	BA		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK3mm INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	10"	2	EA	
1264	AREA PRODUKSI	PL-064-BA-10"	TMBP-01-PR-PID-107	GASKET	BA		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK3mm INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	10"	2	EA	
1265	AREA PRODUKSI	PL-065-BA-4"	TMBP-01-PR-PID-107	PIPE	BA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	17,213	M	
1266	AREA PRODUKSI	PL-065-BA-4"	TMBP-01-PR-PID-107	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
1267	AREA PRODUKSI	PL-065-BA-4"	TMBP-01-PR-PID-107	90 ELBOW	BA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	6	EA	
1268	AREA PRODUKSI	PL-065-BA-4"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL 300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	5	EA	
1269	AREA PRODUKSI	PL-065-BA-4"	TMBP-01-PR-PID-107	FLANGE	BA	40	300	WN-RF	FLANGE, A105, CL 300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	1	EA	
1270	AREA PRODUKSI	PL-065-BA-4"	TMBP-01-PR-PID-107	EQUAL TEE	BA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
1271	AREA PRODUKSI	PL-065-BA-4"	TMBP-01-PR-PID-107	BOLT&NUTS	BA		300		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x115mm	40	EA	
1272	AREA PRODUKSI	PL-065-BA-4"	TMBP-01-PR-PID-107	GASKET	BA		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK3mm INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	4	EA	
1273	AREA PRODUKSI	PL-065-BA-4"	TMBP-01-PR-PID-107	GASKET	BA		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK3mm INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	2	EA	
1274	AREA PRODUKSI	PL-070-AA-3"	TMBP-01-PR-PID-104	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	27,388	M	
1275	AREA PRODUKSI	PL-070-AA-3"	TMBP-01-PR-PID-104	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	6	EA	
1276	AREA PRODUKSI	PL-070-AA-3"	TMBP-01-PR-PID-104	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
1277	AREA PRODUKSI	PL-070-AA-3"	TMBP-01-PR-PID-104	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
1278	AREA PRODUKSI	PL-070-AA-3"	TMBP-01-PR-PID-104	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	16	EA	
1279	AREA PRODUKSI	PL-070-AA-3"	TMBP-01-PR-PID-104	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	4	EA	
1280	AREA PRODUKSI	PL-071-AA-20"	TMBP-01-PR-PID-104	PIPE	AA	40	150	BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	20"	5,711	M	
1281	AREA PRODUKSI	PL-071-AA-20"	TMBP-01-PR-PID-104	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	20"	2	EA	
1282	AREA PRODUKSI	PL-071-AA-20"	TMBP-01-PR-PID-104	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	20"	2	EA	
1283	AREA PRODUKSI	PL-071-AA-20"	TMBP-01-PR-PID-104	BOLT&NUTS	AA		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	1 1/8"x160mm	40	EA	
1284	AREA PRODUKSI	PL-071-AA-20"	TMBP-01-PR-PID-104	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	20"	2	EA	
1285	AREA PRODUKSI	PL-073-AA-8"	TMBP-01-PR-PID-105	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	8"	14,21	M	
1286	AREA PRODUKSI	PL-073-AA-8"	TMBP-01-PR-PID-105	90 ELBOW	AA	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	1	EA	
1287	AREA PRODUKSI	PL-073-AA-8"	TMBP-01-PR-PID-105	BLIND FLANGE	AA		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	8"	1	EA	
1288	AREA PRODUKSI	PL-073-AA-8"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	2	EA	
1289	AREA PRODUKSI	PL-073-AA-8"	TMBP-01-PR-PID-105	EQUAL TEE	AA	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	2	EA	
1290	AREA PRODUKSI	PL-073-AA-8"	TMBP-01-PR-PID-105	BOLT&NUTS	AA		300		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	24	EA	
1291	AREA PRODUKSI	PL-073-AA-8"	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	3	EA	
1292	AREA PRODUKSI	PL-074-AA-8"	TMBP-01-PR-PID-105	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	8"	5,777	M	
1293	AREA PRODUKSI	PL-074-AA-8"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	1	EA	
1294	AREA PRODUKSI	PL-074-AA-8"	TMBP-01-PR-PID-105	BOLT&NUTS	AA		300		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	8	EA	
1295	AREA PRODUKSI	PL-074-AA-8"	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	1	EA	
1296	AREA PRODUKSI	PL-075-AA-8"	TMBP-01-PR-PID-105	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	8"	3,683	M	
1297	AREA PRODUKSI	PL-075-AA-8"	TMBP-01-PR-PID-105	45 ELBOW	AA	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	1	EA	
1298	AREA PRODUKSI	PL-075-AA-8"	TMBP-01-PR-PID-105	FLANGE	AA	40	150	WN-RF	FLANGE, A105, CL 150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	1	EA	
1299	AREA PRODUKSI	PL-075-AA-8"	TMBP-01-PR-PID-105	BOLT&NUTS	AA		300		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	8	EA	
1300	AREA PRODUKSI	PL-075-AA-8"	TMBP-01-PR-PID-105	GASKET	AA		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	1	EA	
1301	AREA UTILITY	T-019_N8	TMBP-01-PR-PID-120	PIPE	AA	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	0,821	M	
1302	AREA UTILITY	T-019_N8	TMBP-01-PR-PID-120	CAP	AA	40		BW	CAP, A234-WPB SMLS, SCH40, BW, ASME B16.9	4"	2	EA	

ATTACHMENT-2 SUMMARY MTO PIPES, FLANGES AND FITTINGS

SUMMARY MTO PIPES, FLANGES AND FITTINGS

JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR

NO.	PART NAME	SCHEDULE	RATING	CONNECTION END	DESCRIPTION	SIZE (Inch)	QTY	UNIT	REMARKS
AREA PRODUKSI									
1. PIPES									
1	PIPE	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	711,00	M	
2	PIPE	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	690,83	M	
3	PIPE	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	742,66	M	
4	PIPE	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	8"	230,98	M	
5	PIPE	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	10"	51,07	M	
6	PIPE	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	20"	5,71	M	
2. FITTINGS									
1	45 ELBOW	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	19	EA	
2	45 ELBOW	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	28	EA	
3	45 ELBOW	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	17	EA	
4	45 ELBOW	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	2	EA	
5	45 ELBOW	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	10"	2	EA	
6	90 ELBOW	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	143	EA	
7	90 ELBOW	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	139	EA	
8	90 ELBOW	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	86	EA	
9	90 ELBOW	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	8	EA	
10	90 ELBOW	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	10"	4	EA	
11	EQUAL TEE	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	103	EA	
12	EQUAL TEE	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	33	EA	
13	EQUAL TEE	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	29	EA	
14	EQUAL TEE	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	2	EA	
15	EQUAL TEE	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	10"	10	EA	
16	REDUCING TEE	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	12	EA	
17	REDUCING TEE	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x3"	30	EA	
18	REDUCING TEE	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	9	EA	
19	CONCENTRIC REDUCER	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	12"x6"	2	EA	
20	CONCENTRIC REDUCER	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	4	EA	
21	CONCENTRIC REDUCER	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	9	EA	
22	ECCENTRIC REDUCER	40		BW	ECC. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	10"x8"	2	EA	
23	ECCENTRIC REDUCER	40		BW	ECC. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	2	EA	
24	ECCENTRIC REDUCER	40		BW	ECC. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x3"	3	EA	
25	CAP	40		BW	CAP, A234-WPB SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
3. FLANGES									
1	FLANGE	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	276	EA	
2	FLANGE	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	10	EA	
3	FLANGE	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	116	EA	
4	FLANGE	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	14	EA	
5	FLANGE	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	110	EA	
6	FLANGE	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	22	EA	
7	FLANGE	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	2	EA	
8	FLANGE	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	7	EA	
9	FLANGE	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	10"	16	EA	
10	FLANGE	40	300	WN-RF	FLANGE, A105, CL.300, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	10"	8	EA	
11	FLANGE	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	12"	2	EA	
12	FLANGE	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	20"	2	EA	
4. BLIND FLANGE									
1	BLIND FLANGE		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	3"	63	EA	
2	BLIND FLANGE		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	20	EA	
3	BLIND FLANGE		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	6"	17	EA	
4	BLIND FLANGE		300	RF	BLIND FLANGE, A105, CL.300, RF, ASME B16.5, SMOOTH FINISHED	6"	2	EA	
5	BLIND FLANGE		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	8"	6	EA	
6	BLIND FLANGE		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	10"	4	EA	
7	BLIND FLANGE		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	16"	6	EA	
8	BLIND FLANGE		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	20"	2	EA	
9	SPECTACLE BLIND		150	RF	SPECTACLE BLIND, A515-70, CL.150, RF, ASME B16.48	3"	3	EA	
10	SPECTACLE BLIND		150	RF	SPECTACLE BLIND, A515-70, CL.150, RF, ASME B16.48	4"	3	EA	
11	SPECTACLE BLIND		150	RF	SPECTACLE BLIND, A515-70, CL.150, RF, ASME B16.48	6"	3	EA	
12	SPECTACLE BLIND		150	RF	SPECTACLE BLIND, A515-70, CL.150, RF, ASME B16.48	12"	2	EA	
5. GASKET									
1	GASKET		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	313	EA	
2	GASKET		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK:3mm INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	10	EA	
3	GASKET		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	125	EA	
4	GASKET		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK:3mm INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	12	EA	
5	GASKET		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	121	EA	
6	GASKET		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK:3mm INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	21	EA	
7	GASKET		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	10	EA	
8	GASKET		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	10"	20	EA	
9	GASKET		300	RF	SPIRAL WOUND GASKET, CL.300, SS304 WITH NON ASB, FILLER THK:3mm INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	10"	8	EA	
10	GASKET		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	12"	4	EA	
11	GASKET		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	20"	2	EA	
6. BOLT & NUT									
1	BOLT&NUTS		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	1 1/8"x160mm	40	EA	
2	BOLT&NUTS		300		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	1"x160mm	128	EA	
5	BOLT&NUTS		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	944	EA	
6	BOLT&NUTS		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	160	EA	
7	BOLT&NUTS		300		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x115mm	88	EA	

NO.	PART NAME	SCHEDULE	RATING	CONNECTION END	DESCRIPTION	SIZE (Inch)	QTY	UNIT	REMARKS
8	BOLT&NUTS		300		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x120mm	240	EA	
9	BOLT&NUTS		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	2256	EA	
10	BOLT&NUTS		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	7/8"x115mm	240	EA	
11	BOLT&NUTS		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	7/8"x120mm	24	EA	
AREA UTILITY									
1. PIPES									
1	PIPE	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	458,68	M	
2	PIPE	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	316,93	M	
3	PIPE	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	479,71	M	
2. FITTINGS									
1	45 ELBOW	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	11	EA	
2	45 ELBOW	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	13	EA	
3	45 ELBOW	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	5	EA	
4	90 ELBOW	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	70	EA	
5	90 ELBOW	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	41	EA	
6	90 ELBOW	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	26	EA	
7	EQUAL TEE	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	8	EA	
8	EQUAL TEE	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	16	EA	
9	EQUAL TEE	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9, GALV.	4"	1	EA	
10	REDUCING TEE	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	3	EA	
11	CONCENTRIC REDUCER	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	10"x6"	1	EA	
12	CONCENTRIC REDUCER	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	1	EA	
13	CONCENTRIC REDUCER	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	5	EA	
14	CONCENTRIC REDUCER	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"x6"	1	EA	
15	ECCENTRIC REDUCER	40		BW	ECC. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	1	EA	
16	CAP	40		BW	CAP, A234-WPB SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
3. FLANGES									
1	FLANGE	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	59	EA	
2	FLANGE	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	39	EA	
3	FLANGE	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	9	EA	
4	FLANGE	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	1	EA	
5	FLANGE	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	10"	1	EA	
4. BLIND FLANGE									
1	BLIND FLANGE		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	3"	2	EA	
2	BLIND FLANGE		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	8	EA	
3	BLIND FLANGE		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	6"	2	EA	
6	SPECTACLE BLIND		150	RF	SPECTACLE BLIND, A515-70, CL.150, RF, ASME B16.48	3"	1	EA	
7	SPECTACLE BLIND		150	RF	SPECTACLE BLIND, A515-70, CL.150, RF, ASME B16.48	8"	1	EA	
8	SPECTACLE BLIND		150	RF	SPECTACLE BLIND, A515-70, CL.150, RF, ASME B16.48	10"	1	EA	
5. GASKET									
1	GASKET		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	52	EA	
2	GASKET		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	40	EA	
3	GASKET		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	6	EA	
4	GASKET		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	2	EA	
5	GASKET		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	10"	1	EA	
6. BOLT & NUT									
1	BOLT&NUTS		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	48	EA	
2	BOLT&NUTS		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	8	EA	
3	BOLT&NUTS		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	524	EA	
AREA FIRE WATER									
1. PIPES									
1	PIPE	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	3"	244,63	M	
2	PIPE	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	4"	103,73	M	
3	PIPE	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	6"	38,00	M	
4	PIPE	40		BE	PIPE, A106-B, SMLS, SCH40, BE, ASME B36.10M	8"	950,30	M	
2. FITTINGS									
1	45 ELBOW	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	6	EA	
2	45 ELBOW	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	1	EA	
3	45 ELBOW	40		BW	45 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	4	EA	
4	90 ELBOW	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	87	EA	
5	90 ELBOW	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	19	EA	
6	90 ELBOW	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"	7	EA	
7	90 ELBOW	40		BW	90 ELBOW, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	62	EA	
9	EQUAL TEE	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	3"	22	EA	
10	EQUAL TEE	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"	2	EA	
11	EQUAL TEE	40		BW	TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"	9	EA	
12	REDUCING TEE	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	5	EA	
13	REDUCING TEE	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	6"x4"	1	EA	
14	REDUCING TEE	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"x4"	13	EA	
15	REDUCING TEE	40		BW	RED. TEE, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"x6"	3	EA	
17	CONCENTRIC REDUCER	40		BW	CON. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	4"x3"	2	EA	
18	ECCENTRIC REDUCER	40		BW	ECC. REDUCER, A234-WPB, SMLS, SCH40, BW, ASME B16.9	8"x6"	1	EA	
19	WELDOLET	40		BW	WELDOLET, A105, SCH40, BW, MSS SP-97	8"x3"	10	EA	
3. FLANGES									
1	FLANGE	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	3"	98	EA	
2	FLANGE	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	4"	20	EA	
3	FLANGE	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	6"	9	EA	
4	FLANGE	40	150	WN-RF	FLANGE, A105, CL.150, SCH40, WN-RF, ASME B16.5, SMOOTH FINISHED	8"	68	EA	

skkmigas / PERTAMINA EP	SUMMARY MTO PIPES, FLANGES AND FITTINGS JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR	prosympac O&G

NO.	PART NAME	SCHEDULE	RATING	CONNECTION END	DESCRIPTION	SIZE (Inch)	QTY	UNIT	REMARKS
4. BLIND FLANGE									
1	BLIND FLANGE		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	3"	4	EA	
2	BLIND FLANGE		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	4"	2	EA	
3	BLIND FLANGE		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	6"	1	EA	
4	BLIND FLANGE		150	RF	BLIND FLANGE, A105, CL.150, RF, ASME B16.5, SMOOTH FINISHED	8"	1	EA	
5. GASKET									
1	GASKET		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	3"	85	EA	
2	GASKET		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	4"	33	EA	
3	GASKET		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	6"	8	EA	
4	GASKET		150	RF	SPIRAL WOUND GASKET, CL.150, SS304 WITH NON ASB, FILLER THK:3MM INNER AND OUTER COMPRESSION RINGS, RF ASME B16.5/ASME B16.20	8"	53	EA	
6. BOLT & NUT									
1	BOLT&NUTS		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x100mm	64	EA	
2	BOLT&NUTS		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	3/4"x110mm	424	EA	
3	BOLT&NUTS		150		STUD BOLT&NUTS, A193-B7, FULL-THD, ASME B18.2.1/B1.1 CL.2A/ A194-2H, HEAVY HEX. NUTS, ASME B18.2.2/B1.1 CL.2B, NYLON COAT OR EQUAL	5/8"x90mm	604	EA	